

ONTARIO

SUPERIOR COURT OF JUSTICE

BETWEEN:

RE: THE ESTATE OF MARY FLEURY,
by Estate Trustee, GEORGE
MCAMMOND, GEORGE MCAMMOND,
DAMIAN REID, a minor by his litigation
guardian, GEORGE MCAMMOND, and
GAVIN BRADFORD, a minor by his
litigation guardian, GEORGE
MCAMMOND

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) Paul Harte, Ron Bohm and Giuseppe
) Michelucci, for the Plaintiffs
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– and –

OLAYIWOLA A. KASSIM

) Erica J. Baron, Charlotte-Anne
) Malischewski and Kate Martini, for the
) Defendant

HEARD:

December 6,7,8,9,10,13,14,15,16,17
and 21, 2021

JUSTICE S.K. STOTHART

I. INTRODUCTION

- [1] On October 31, 2015, Mary Fleury underwent a laparoscopy at the West Parry Sound Health Centre for a suspected small bowel obstruction. During the operation the surgeon noticed evidence suggestive of cancer in Mary's abdomen. Further testing following surgery revealed that Mary had metastatic adenocarcinoma of the appendix.
- [2] On November 26, 2016, a little over a year later, Mary died from complications related to that cancer.

- [3] The Plaintiffs allege that Dr. Olayiwola Kassim, an anatomic pathologist, was negligent when he failed to identify and diagnose Mary Fleury's appendiceal cancer when she had her appendix removed in 2011. The Plaintiffs further allege that had Mary been properly diagnosed in 2011, she would be alive today.
- [4] The Defendant says that he made an honest mistake when he failed to identify Ms. Fleury's cancer in 2011, and that he was not negligent. The Defendant further says that even if he had diagnosed Mary Fleury's cancer in 2011, she would have regrettably died in any event.
- [5] For the following reasons this court concludes, on a balance of probabilities, that Dr. Kassim did not meet the requisite standard of care when he examined Mary Fleury's appendiceal samples in 2011 and failed to identify and diagnose cancer.
- [6] Further this court concludes that, had Mary been properly diagnosed, she would likely be alive today. As such, her estate, her spouse and her two grandchildren are entitled to damages.

II. BACKGROUND:

- [7] In October, 2011, Mary Fleury developed appendicitis and underwent a routine appendectomy at the West Parry Sound Health Centre. The surgeon removed Mary's appendix and it was sent to the pathology department where samples were examined by Dr. Olayiwola Kassim, who was the Head of the Pathology Department at the West Parry Sound Health Centre.
- [8] On October 26, 2011, Dr. Kassim reported his findings as "perforated acute appendicitis with periappendiceal abscess". Dr. Kassim noted a ruptured inflamed appendix, which had been obstructed by fecal matter with secondary acute purulent inflammation and a periappendiceal abscess. Dr. Kassim made no note or report of any malignancy.
- [9] In 2015, Mary began to suffer from bloating and pain on the right side of her abdomen (right lower quadrant). From June to October, 2015, Mary presented at the West Parry Sound Health Centre several times with this concern. This led to testing including x-rays, CT scans, a colonoscopy and an esophageal gastroduodenoscopy.
- [10] Doctors began to suspect a small bowel obstruction with ascites (fluid in the peritoneal cavity). On October 31, 2015, Mary underwent a laparoscopy. It was during this procedure that the surgeon noticed diffuse seeding of the parietal peritoneum which was suggestive of metastatic disease. The surgeon took samples of the peritoneal tissue and ascitic abdominal fluid and sent them to the pathology lab for examination.

- [11] Shortly after the surgery, the surgeon spoke to Mary's spouse George McAmmond and told him that they had found cancer. When George asked how much cancer, he was told it was a lot and that the focus would be on improving Mary's remaining quality of life, rather than curing her. Mary, George and her family were understandably devastated by this news.
- [12] On November 9, 2015, Dr. Kassim examined the tissue samples and the ascitic fluid taken by the surgeon and found atypical glandular and spindle cell proliferation, which is a potential indicator of cancer.
- [13] The samples were then sent to the gastrointestinal pathology service of Mount Sinai Hospital, who reported that the biopsies exhibited "findings consistent with metastatic adenocarcinoma with the possibility of spread from a gastrointestinal-tract primary carcinoma".
- [14] On November 12, 2015, Mary underwent a right hemicolectomy with resection of her terminal ileum, cecum, a portion of the omentum and 11 lymph nodes. Dr. Kassim examined the tissue removed during this procedure and found a tumour mass infiltrating the wall of the terminal ileum, which he diagnosed as an adenocarcinoma ex goblet cell carcinoid tumour. The tumour encompassed the lower end of the cecum, where the appendix should have been. Dr. Kassim also diagnosed cancer in the omentum. He did not find any evidence of cancer in the lymph nodes.
- [15] After Mary was diagnosed with cancer, doctors began investigating where the cancer had originated from. Dr. Kassim contacted Soldiers' Memorial Hospital, where Mary had had a hysterectomy and oophorectomy in 2011. He wanted to see whether the cancer originated in the uterus or ovaries. When Dr. Kassim examined the slides taken from those procedures he found no evidence of cancer.
- [16] Dr. Kassim also made inquiries about whether Mary had ever had her appendix removed because during his examination of the samples from the hemicolectomy he did not observe the appendix. Dr. Kassim wasn't sure if the appendix had been consumed by the tumour, or whether it had been removed previously.
- [17] Dr. Kassim discovered that Mary had her appendix removed in 2011 at the West Parry Sound Health Centre. Further, that he was the pathologist who examined Mary's appendiceal samples at that time. When he realized this, he ordered the pathology slides from the basement of the hospital and re-examined them.
- [18] During his re-examination of the 2011 appendiceal slides Dr. Kassim observed that the appendiceal samples contained cancer. On November 25, 2015, Dr. Kassim amended his original 2011 diagnosis and replaced it with "infiltrating adenocarcinoma ex goblet cell tumour of the appendix with ruptured acute on chronic appendicitis with marked diffuse fibrosis". He staged the tumour as pT4a – which meant the tumour had invaded through the appendiceal wall. This was described variously by witnesses as meaning that the cancer had spread to the outside surface of the appendix.

- [19] Dr. Kassim sent the original 3 slides from 2011, along with additional slides he created from the appendiceal specimen in 2015, to Mount Sinai's gastrointestinal pathology consultation service for review. The consensus opinion of the three reviewing pathologists was that the specimens contained goblet cell carcinoma with a small component of adenocarcinoma ex GCC, involving the serosal surface with radial margin positive for GCC.
- [20] On December 18, 2015, Mary Fleury attended at Princess Margaret Hospital for a consultation with the oncology service. The plan was to start chemotherapy and reattend three months later for re-assessment. Mary underwent three rounds of chemotherapy between February and March, 2016 but experienced toxic side effects and stopped the therapy.
- [21] On August 23, 2016, Mary was reassessed by her oncologist who concluded that her cancer had progressed. Further chemotherapy was offered, but Mary declined due to the severe side effects. Mary opted for improved quality of life for the little time she had left.
- [22] On October 4, 2016, Mary was admitted to the West Parry Sound Health Centre for palliative care. She died on November 26, 2016, at the age of 45.

III. THE POSITION OF THE PARTIES

A. The Plaintiffs' position

- [23] The Plaintiffs argue that the Defendant, Dr. Kassim, was negligent when he failed to identify, diagnose, and communicate to Mary Fleury, that she had cancer in 2011. They argue that Dr. Kassim failed to exercise the reasonable degree of skill, knowledge and care expected of an anatomic pathologist in his circumstances when he reviewed Mary Fleury's 2011 appendiceal samples and failed to identify and diagnose the cancer that was obviously there.
- [24] The Plaintiffs further argue that but for Dr. Kassim's failure to exercise the standard of care expected of an anatomic pathologist in similar circumstances, Mary Fleury would likely have sought out treatment for cancer and would be alive today.

B. The Defendant's position

- [25] The Defendant argues that the Plaintiffs have failed to demonstrate, on a balance of probabilities, that any act or omission of Dr. Kassim fell below the standard of care of a reasonable pathologist in similar circumstances. The Defendant argues that a physician is not held to a standard of perfection and an honest and intelligent exercise of judgement satisfies the standard of care. An error in judgement, alone, does not amount to negligence.

- [26] The Defendant further argues that even if the court concludes that the Plaintiffs have demonstrated a breach of the standard of care, that the Plaintiffs have failed to demonstrate that Ms. Fleury's cancer was likely capable of cure in October, 2011.

IV. THE ISSUES IN THIS CASE

- [27] The required elements that must be proven, on a balance of probabilities, in any case alleging the tort of negligence were set out by the Supreme Court of Canada in *Mustapha v. Culligan of Canada Ltd.*, 2008 SCC 27 at para 3. The elements are: that the Defendant owed a duty of care, that the Defendant's behaviour breached the standard of care, that the Plaintiff sustained damage, and that the damage was caused, in fact and in law, by the Defendant's breach.

- [28] It is agreed between the parties that the Defendant owed Mary Fleury a duty of care when he examined her appendiceal samples as a pathologist in 2011. The parties agree that there are two central issues in this case. They are:

1. **Whether there was a breach of the standard of care.** Specifically, whether the Plaintiffs have established, on a balance of probabilities, that Dr. Kassim fell below the standard of care expected of a similarly situated anatomic pathologist, when he failed to identify the cancer present in Mary Fleury's appendix in October, 2011. And if so, how did he fall below this standard; and
2. **Causation.** If the court concludes that Dr. Kassim breached the standard of care expected of him, whether the Plaintiffs have established, on a balance of probabilities, that this breach caused or contributed to the death of Mary Fleury.

V. STANDARD OF CARE

- [29] The standard of care required of a medical practitioner is to exercise a reasonable degree of skill and knowledge and the degree of care that could reasonably be expected of a normal, prudent practitioner of the same experience and standing. If the medical practitioner holds him/herself out as a specialist, a higher degree of skill is required as a result of this special training and ability. *Crits v. Sylvester* [1956] O.J. No. 526 (ONCA) aff'd [1956] SCR 991. *Armstrong v. Royal Victoria Hospital*, 2019 ONCA 963 at para 86, aff'd 2021 SCC 1.

- [30] The standard of reasonableness is not a standard of excellence that amounts to perfection. A physician's honest and intelligent exercise of judgement will satisfy the standard of care. Physicians are not held to a standard of perfection and are not liable for errors in judgement. Rather, the standard expected of a medical professional

must be realistic and reasonable. *Armstrong v. Royal Victoria Hospital, supra.* at para 86; *Tacknyk v. Lake of the Woods Clinic*, [1982] O.J. No. 170 (Ont.C.A.) at para. 29.

[31] In a case of alleged medical malpractice, the court must determine what was reasonably required to be done (or avoided) by the Defendant physician in order to meet the standard of care. *Armstrong v. Royal Victoria Hospital, supra* at para. 86.

[32] In reviewing what was reasonably required to be done (or avoided) by a medical practitioner, the court must be careful to ensure that it assesses the physician's conduct in light of the medical knowledge and circumstances that they ought to have been aware of at the time of the alleged negligence, without the benefit of hindsight. *ter Neuzen v. Koran*, [1995] 3 S.C.R. 674 at para. 33.

[33] Expert evidence is often necessary in assisting the trier of fact in determining whether or not a Defendant has breached the standard of care. In *Bafaro v. Dowd*, [2008] O.J. No. 3474 (Ont.S.C.J.) aff'd (2010), 260 O.A.C. 70 (Ont.C.A.), the court held at para. 31 that:

Actions alleging malpractice involve issues to be decided that are not within the ordinary knowledge and experience of the trier of fact. Therefore, the Plaintiff requires expert evidence to prove that the Defendant physician was negligent.

[34] In this case the court is tasked with determining whether the Plaintiffs have shown, on a balance of probabilities, that the cancer in Mary Fleury's appendiceal slides should have been seen using standard pathologic practice by a reasonably prudent pathologist. *Armstrong v. Royal Victoria Hospital, supra.* at para 120.

A. The work of an anatomic pathologist

[35] Pathology was described at the trial as the art and science of the abnormal in human diseases. It requires specialized training and experience.

[36] The experts who testified at trial described a pathologist's work as including the examination of specimens received from a patient from surgery, a biopsy or other procedure. When a pathologist examines a specimen, they are looking for anything abnormal, such as infections, malignancies, or genetic disorders.

[37] All of the pathologists who testified at trial agreed that a pathologist is always looking for cancer and that this is a standard feature of their work.

[38] With respect to the standard process followed by an anatomic pathologist when examining a tissue sample from a patient, all of the pathologists agreed that the following process takes place:

1. A specimen is taken by a medical practitioner, usually as part of surgery or a biopsy procedure;
2. This specimen is brought to a pathology lab along with a requisition that contains patient information, the name of the doctor who sent the specimen, the surgery performed, what was retrieved from the body and a clinical diagnosis;
3. A pathology assistant conducts a gross examination and notes their observations;
4. The specimen is then processed by the lab and ultimately placed into a block of wax;
5. The block of wax is then cut into thin slices. The thin slices are touched on to a glass slide, stained with a standard stain and then covered;
6. That slide, along with paperwork, is delivered to the pathologist for examination;
7. When a slide is received by an anatomic pathologist, they review the paperwork, including the requisition and noted gross observations;
8. The pathologist then looks at the glass slide using a microscope to formulate a diagnosis. This process starts with scanning the entire slide under low magnification, and then moving to higher levels of magnification to examine any areas of concern; and
9. The pathologist then dictates a report containing their diagnosis which is then distributed back to the referring physician and medical records.

B. Expert evidence with respect to the standard of care expected of an anatomic pathologist

Dr. Nihad Hadi Ali-Ridha

[39] The Plaintiffs called Dr. Nihad Hadi Ali-Ridha, who was qualified to provide opinion evidence as to the standard of care in Ontario with respect to pathology in general and specifically cancer pathology, including the preparation of specimens and the review and reporting of pathologic findings.

Qualifications

- [40] Dr. Ali-Ridha is an anatomic pathologist who currently works at Lakeridge Health Corporation in Oshawa, Ontario. He is also a urological pathology consultant with the Regional Durham Cancer Centre, an associate medical staff at Rouge Valley Health Centre, and is a locum pathologist for the Southlake Health Centre.
- [41] Dr. Ali-Ridha has been qualified to practice as a pathologist in Canada since 1988. During his career he has trained and practiced in Alberta, Nova Scotia, New Brunswick and Ontario. He currently has a clinical academic appointment with the Department of Laboratory Medicine and Pathobiology with the Faculty of Medicine at the University of Toronto and has taught medical, dental and 3rd to 4th year pathology residents at the University of Toronto for 21 years.
- [42] During his career, Dr. Ali-Ridha has worked at small and large hospitals throughout Ontario, with the majority of his career being spent at larger hospitals. He has particular expertise in gastro-intestinal and genital-urinary cancer and sub-specializes in this area.

Evidence of Dr. Ali-Ridha on the standard of care

- [43] In preparation for his opinion on the issue of standard of care, Dr. Ali-Ridha reviewed the pathology reports, glass slides of samples taken from Mary Fleury in both 2011 and 2015, Dr. Lage's report and the discovery transcript of Dr. Kassim.
- [44] It was Dr. Ali-Ridha's opinion, having reviewed these materials, that Dr. Kassim did not meet the standard of care expected of a competent pathologist in Ontario when Dr. Kassim reviewed Mary Fleury's appendiceal samples in October, 2011.
- [45] Dr. Ali-Ridha testified that Dr. Kassim should have been able to identify the cancer in Mary Fleury's appendiceal samples using a simple microscope, at low magnification, without resort to any extraordinary techniques. Dr. Ali-Ridha testified "any pathologist with basic training would have called it" and that the cancer was obvious.
- [46] During his review, Dr. Ali-Ridha took high-quality digital photographs of Mary Fleury's 2011 slides. One of these digital photographs was tendered at trial and marked as Exhibit 5. Dr. Ali-Ridha described the photograph as depicting one of Mary Fleury's appendiceal slides viewed under a microscope at the lowest scanning power or screening power. This is the view that a pathologist would have when they first examine the slide under a microscope, before going on to further examine the slide at higher resolution.
- [47] Dr. Ali-Ridha testified that he added green boxes to Exhibit 5, to highlight the most obvious areas of cancer in that slide. The green boxes show the areas of cancer that had better morphology, where one would not have to use a higher resolution to diagnose the cancer.

- [48] Dr. Ali-Ridha testified that the inflammation present Mary Fleury's 2011 appendiceal sample would not be a valid reason for a pathologist, exercising the appropriate standard of care, to fail to observe the cancer present in the slides. He testified that while inflammation can make the job more difficult, an experienced pathologist would recognize that the pattern of white cell infiltration in the tissue is different than cancer cells. He was not challenged on this opinion in cross-examination.
- [49] In cross-examination, Dr. Ali-Ridha was challenged on the fact that he had prior knowledge of a missed cancer diagnosis when he reviewed the slides. It was suggested that his evidence was tainted by hindsight.
- [50] Dr. Ali-Ridha was not cross-examined on the issue of whether the 2011 slides were overstained and whether that could impact a pathologist's ability to observe the cancer contained in the slides, notwithstanding that this was a significant point raised by the Defendant's expert.
- [51] It was pointed out during cross-examination that when Dr. Ali-Ridha examined Mary Fleury's 2015 lymph nodes, he observed cancer in five lymph nodes contrary to Dr. Kassim's evidence that he saw no evidence of cancer in any of Mary Fleury's 2015 lymph nodes. This will become relevant later on in the court's analysis on causation.
- [52] There were times when Dr. Ali-Ridha's responses in cross-examination were not responsive to the question asked. The court observed that sometimes this was due to the use of the Zoom platform and poor sound quality. At other times it was apparent that Dr. Ali-Ridha had either not heard or understood the question being asked before answering it. Having reviewed his evidence in its entirety, the court concludes that Dr. Ali-Ridha's evidence was fair, objective and reliable. The court did not find the substance of his evidence to be significantly impacted by cross-examination.
- [53] There will always be an aspect of hindsight bias in any case where an expert physician has been asked to opine on the actions of another physician. The expert has been retained by a lawyer and would naturally infer that there must be some issue with the physician's conduct.
- [54] Because of this inherent bias, the court is required to assess the expert's evidence to determine whether the expert has held the physician to a higher standard than would have been expected of the physician in light of the medical knowledge and circumstances that they ought to have been aware of at the time of the alleged negligence.
- [55] In this case, the court concludes that Dr. Ali-Ridha did not hold Dr. Kassim's conduct to a higher standard than what would be expected. This isn't a case where, with the knowledge that Mary Fleury had cancer, Dr. Ali-Ridha took extraordinary steps to try to find the cancer. The cancer was visible at low magnification, which would be the first step that a pathologist takes when reviewing a slide.

C. Expert evidence with respect to the interpretation of surgical slides

Dr. Janice Lage

- [56] The Defendant called Dr. Janice Lage, who was qualified to provide expert opinion with respect to the interpretation of surgical slides including the ways in which diagnostic discrepancies in interpretation can occur.
- [57] Dr. Lage was also qualified to provide expert evidence with respect to the interpretation and classification of the surgical slides from Mary Fleury's procedures as it related to the likely spread of Ms. Fleury's cancer in 2011.
- [58] Dr. Lage was not qualified as an expert with respect to the standard of care expected of an anatomic pathologist.

Qualifications

- [59] Dr. Lage is an anatomic pathologist, currently working at St. Michael's hospital in Toronto, Ontario as a surgical pathologist. Although Dr. Lage sub-specializes in obstetric, gynecological and placental pathology, she does possess training and experience with respect to the review of appendiceal slides.
- [60] Dr. Lage has held supervisory positions during her career where she oversaw the work of other pathologists, including reviewing their errors and looking at ways to decrease error rates.

Evidence of Dr. Lage with respect to pathology slides

- [61] Dr. Lage provided both a specific opinion about Mary Fleury's 2011 slides, and a general opinion on factors that could contribute to an error in diagnosis.
- [62] Dr. Lage was not asked to opine on the standard of care expected of an anatomic pathologist. As such, she did not review the evidence of Dr. Kassim either at discovery or at trial.
- [63] With respect to Mary Fleury's 2011 slides, it was Dr. Lage's opinion that the slides were overstained. She testified that when she reviewed the slides, she could not distinguish the cancer cells from normal cells as readily as she could have had the slides been appropriately stained. It was her opinion that the overstaining could have caused a pathologist's eye to be drawn to the area of the inflammation and the location where the appendix ruptured.
- [64] With respect to factors that could lead to a misdiagnosis generally, Dr. Lage testified that:
1. Inflammation, related to appendicitis, can obscure what a pathologist sees;

2. A pathologist's "index of suspicion" for cancer could be lowered if there is no clinical concern related to cancer communicated to the pathologist; and
3. Errors can be caught by a referring surgeon questioning a pathologist's report or asking for a second review.

[65] Dr. Lage testified that in her practice, when she receives a slide from a lawyer she knows that there is a problem and that it is her obligation to keep looking at the slide until she finds the problem. She described this as a diagnosis bias.

[66] Dr. Lage testified that one way to avoid diagnosis bias is to re-label the slide and put it into the daily work of a pathologist and see if they pick out the problem. This gives a better idea of what a pathologist would have seen in a daily review.

[67] Dr. Lage's evidence with respect to "misdiagnosis" or "errors" was relevant to whether there were non-negligent ways that the cancer could have been missed. Her evidence was relevant to whether Dr. Kassim's failure to diagnose Mary Fleury's cancer fell within the exercise of a "reasonable degree of skill and knowledge" by a normal, prudent pathologist as opposed to holding Dr. Kassim to a standard of perfection.

[68] In my view, Dr. Lage's explanations in support of the reasonableness of Dr. Kassim's failure to diagnose Mary Fleury's cancer in 2011 were undermined by Dr. Kassim's own evidence.

[69] Dr. Lage's opinion that the slides were overstained was contradicted by Dr. Kassim who testified that he felt the staining was sufficient for him to appropriately review the slides both in 2011 and 2015. He testified that it would have been his responsibility to ensure that the staining was appropriate, and if the staining was not suitable for examination he would have asked for further slides to be prepared. This may explain why Dr. Ali-Ridha was not cross-examined on this point, because it did not assist the Defendant.

[70] With respect to Dr. Lage's opinion related to index of suspicion, Dr. Kassim testified that he does not initially review the referring physician's clinical diagnosis because he recognizes that his opinion must stand alone and should not be impacted by the clinical impressions.

[71] With respect to Dr. Lage's opinion related to errors being caught by a referring surgeon, Dr. Kassim testified that he does not rely on the referring surgeon to be his quality control. Indeed, it is the role of a pathologist to look for the abnormal and unexpected. That is their specialty and why samples are referred to them for microscopic examination.

[72] This leaves the issue of inflammation and its impact on Dr. Kassim's review of Mary's pathology slides and his diagnosis. With respect to the role inflammation

played in this case, the court was concerned that Dr. Lage seemed unable or unwilling to say whether she, herself, found it difficult to see the cancer when she reviewed the 2011 slides. When specifically asked, Dr. Lage appeared to distance herself from the answer and responded that what she was saying was that it would be difficult or challenging for the original pathologist to see the cancer in those slides. When asked in cross-examination whether she found it moderately easy to see the cancer in the slides, she said she didn't use the term "moderately easy" so didn't know what counsel meant. The court found her evidence to be evasive on this point.

[73] With respect to the issue of diagnosis bias, or hindsight bias, the court found Dr. Lage's evidence to reflect what trial courts have already been directed to do. That is, to review the conduct of a physician in light of the medical knowledge and circumstances that they ought to have been aware of at the time of the alleged negligence, without the benefit of hindsight. In this case, the court specifically turned its mind to this issue when assessing the evidence of Dr. Ali-Ridha.

[74] Overall, the court found Dr. Lage's evidence to be unhelpful when assessed against the totality of evidence called at trial, and Dr. Kassim's evidence in particular. She was, at times, evasive on important points. She would often attribute her difficulty in answering questions to the fact that she had not testified in Canadian proceedings before and so felt a lack of confidence in what she could or could not say.

D. Evidence of Dr. Kassim with respect to the standard of care

[75] Dr. Kassim testified that as an anatomic pathologist, he had certain duties that were expected of him when reviewing Mary Fleury's slides. These included:

1. It was his obligation to ensure that Mary Fleury's appendiceal slides were adequate in all respects in order to conduct a full and proper review;
2. It was his responsibility to approve or not approve the staining of Mary Fleury's slides. If he was not satisfied with the staining, he would ask for additional slides with appropriate staining;
3. It was his responsibility to take the time to carefully review Mary Fleury's pathologic samples to the best of his experience and knowledge; and
4. It was his responsibility to diagnose any observed abnormality and to report that diagnosis back to the reviewing physician.

[76] Dr. Kassim agreed that in fulfilling his responsibilities to Mary Fleury in 2011, he understood that he was required to confirm or disprove the diagnosis of the referring

surgeon. His role was to try to find what others were not suspecting and to set out his relevant findings in his report.

Dr. Kassim's examination of Mary Fleury's appendiceal slides in 2011

- [77] Dr. Kassim could not recall what he did or did not do when he reviewed Mary Fleury's appendiceal slides in 2011. This is understandable given the passage of time between 2011 and 2015.
- [78] Dr. Kassim described the normal process used at his lab in the preparation of sample slides. He testified that there are various quality control measures in place throughout this process before the slide comes to his office for review.
- [79] Relying on his normal practice, he testified that he must have been satisfied with the suitability of Mary Fleury's slides, because he would have sent them back if they were insufficient. He testified that there were no issues with the equipment he was using in 2011.
- [80] Dr. Kassim testified that it is his normal practice to first look at a slide. He then takes the slide and places it under the microscope and slowly moves it from left to right, while also moving it up and down so that he sees the entire tissue. He then repeats this process at increasing levels of magnification.
- [81] Dr. Kassim testified that he does not usually look at the pre-operative diagnosis prior to his examination. This is because this information can be misleading and send a pathologist off in the wrong direction. A pathologist's opinion must stand alone.
- [82] Dr. Kassim testified that in 2015, when he went back to look at the original 2011 slides, he was able to diagnose the cancer present in the slide with moderate ease. He attributed this to the fact that in 2015 he was looking at the slides with knowledge of the cancer diagnosis.
- [83] Dr. Kassim testified that while he could have used more contrast in the staining of Mary Fleury's 2011 slides, he felt the staining was sufficient to make a diagnosis.
- [84] Dr. Kassim testified that he now believes he did not identify Mary Fleury's cancer in 2011 due to the inflammation contained in the specimen and because appendiceal cancer is rare.
- [85] In cross-examination, Dr. Kassim agreed that it is common to see inflammation in a patient that has had appendicitis. The "itis" part of appendicitis means "inflammation".
- [86] Dr. Kassim agreed that when he examines an inflamed appendix, that cancer is always a possible differential diagnosis.

[87] In cross-examination, Dr. Kassim agreed that he had previously diagnosed appendiceal cancer, including goblet cell appendiceal cancer. He had not previously diagnosed goblet cell appendiceal cancer that had an adenocarcinoma component, which is the type of cancer that Mary had.

Dr. Kassim's experience as an anatomic pathologist in 2011

[88] By October, 2011, Dr. Kassim had been employed by the West Parry Sound Health Centre as the Chief of Pathology and Director of Laboratory Services for 26 years. He was an experienced anatomical pathologist who worked primarily in a small community hospital and was a consultant pathologist for hospitals in Orillia, Bracebridge and Huntsville. He was the director of laboratory services for Parry Sound, Elliot Lake and Blind River. He was also an Assistant Professor of pathology and laboratory medicine with the Northern Ontario School of Medicine.

[89] Dr. Kassim testified he participates in ongoing education and has done so throughout his career. This has included attending medical conferences and self-study through the review of academic journals. He testified that he focuses his education on the type of work that he does in a small community hospital. He testified that in his work he performs a lot of gastro-intestinal work and as a result he attends subspecialty conferences in that area.

[90] Dr. Kassim testified that he routinely examines appendiceal specimens and estimated that around 2011 he would see appendiceal specimens about two to three times a week.

[91] Based on the evidence of Dr. Kassim, this court concludes that he is an experienced anatomic pathologist, who has knowledge and experience in gastro-intestinal pathology and experience examining appendiceal specimens including those taken from patients with appendicitis. He would have had this level of experience and training in 2011.

E. Analysis on the Standard of Care

[92] Having considered the evidence in its totality, the court concludes, on a balance of probabilities, that the standard of care expected of an anatomic pathologist includes the following:

1. An anatomic pathologist must ensure that any specimen slide they are examining has been prepared appropriately and is suitable for examination;
2. If the pathologist determines that a slide is unsuitable for examination, they are expected to request that further slides be prepared;

3. Once they receive a slide, they are expected to carefully review the slide to determine if they observe any abnormality, including evidence of cancer;
4. This careful examination is expected to include a gross visual inspection, a general inspection at a low level of magnification and then a closer inspection at higher levels of magnification; and
5. Where a pathologist identifies an abnormality, including possible cancer, they are expected to investigate further and provide a diagnosis based on their findings.

[93] Having considered the evidence called at trial, in its totality, the court is satisfied on a balance of probabilities that Dr. Kassim did not meet the standard of care expected of an anatomic pathologist when he reviewed Mary Fleury's 2011 slides. Specifically, the court is satisfied on a balance of probabilities that Dr. Kassim did not examine Mary Fleury's 2011 slides with the care and expertise expected of a normal, prudent anatomic pathologist with the same level of experience and standing.

[94] The court accepts Dr. Ali-Ridha's evidence that the cancer contained in Mary Fleury's slide was obvious at a low level of magnification. His evidence was supported by Exhibit 5, the digital photograph taken at the lowest level of magnification.

[95] Dr. Ali-Ridha's evidence with respect to the standard of care was not shaken in cross-examination, nor was it contradicted by other expert evidence on this issue.

[96] The evidence of Dr. Kassim was that when he went back to look at the slide in 2015, it was moderately easy to see the cancer. He attributes this to the fact that he already knew about the cancer diagnosis, and so was looking for it. However, he also agreed that a pathologist is always looking for cancer.

[97] The court has turned its mind to whether there were non-negligent ways that the cancer could have been missed in 2011. Specifically, the court has considered the evidence of Dr. Lage and Dr. Kassim on the issues of overstaining and inflammation.

[98] As discussed earlier, Dr. Kassim's testimony does not support the theory that overstaining probably contributed to his failure to observe and diagnose the cancer in the 2011 slides. Ultimately the level of staining did not impair Dr. Kassim, Dr. Ali Ridha or Dr. Lage from observing, diagnosing and staging Mary Fleury's cancer.

[99] With respect to the issue of inflammation, the court accepts the evidence of Dr. Ali-Ridha that the inflammation contained in Mary Fleury's slide would not have impaired the ability of a pathologist, exercising the proper standard of care, from observing the cancer. Appendiceal samples, taken following an appendectomy, necessarily contain inflammation. As Dr. Ali-Ridha explained, an experienced pathologist would recognize that the pattern of white cell infiltration in the tissue is different than cancer cells even

at a low level of magnification. Dr. Ali-Ridha's evidence on this point was not challenged in cross-examination.

[100] The court finds that Dr. Lage's evidence on this issue unhelpful. It was concerning to the court that she would not answer the question put to her in cross-examination, more than once, about whether the inflammation caused her difficulty in examining Mary Fleury's slides. The court found her evidence to be evasive on this point.

[101] It was also telling that Dr. Kassim described being able to identify the cancer with "moderate ease" upon subsequent review, notwithstanding his attempts to qualify this evidence with references to hindsight and the possible impact of inflammation on his review of the slides. In my view, this evidence supported the opinion of Dr. Al-Ridha that Ms. Fleury's cancer was and should have been obvious even to a pathologist with basic training, and certainly to a pathologist with Dr. Kassim's years of practice and experience.

[102] I also find that Dr. Kassim's failure to identify Ms. Fleury's cancer in 2011 was not an error made during an exercise of clinical judgement. I am satisfied on the evidence that the presence of cancer on the pathology slides was obvious and not a subject interpretation or debate. It would have been obvious to a pathologist with Dr. Kassim's skill and experience. His failure to make that finding was a breach of the standard of care.

[103] The court accepts that Dr. Kassim has genuine and profound regret that he did not diagnose Mary Fleury's cancer in 2011. The court further accepts that Dr. Kassim now believes that inflammation played a role in this case. However, the court finds that this is an after the fact justification when faced with the harsh reality that he missed something that he should have seen, which must be very difficult for him.

[104] Some suggestion was made at trial that that the missed diagnosis may have occurred because appendiceal cancer is rare. While the court accepts that this type of cancer is rare, there was no evidence called at trial that linked how rare the cancer was to how difficult it is to see under a microscope. Given that a pathologist is "always looking for cancer" and Dr. Ali-Ridha's evidence that there was no difficulty seeing the cancer in Mary Fleury's appendiceal slide, the court finds that this was not a factor in this case.

VI. CAUSATION

[105] Having found that the Defendant breached the standard of care expected of a normal, prudent anatomic pathologist of the same experience and standing, the court must go on to examine whether the Plaintiffs have established, on a balance of probabilities, that this breach caused the Plaintiffs' injuries. This is often referred to as the "but for" test which requires the Plaintiff to show, on a balance of probabilities, that "but for" the defendant's negligent act, the injury would not have occurred.

[106] The general principles relating to causation were summarized by Justice Major in *Athey v. Leonati* at para. 13-15 as follows:

Causation is established where the plaintiff proves to the civil standard on a balance of probabilities that the defendant caused or contributed to the injury: *Snell v. Farrell*, [1990] 2 S.C.R. 311; *McGhee v. National Coal Board*, [1972] 3 All E.R. 1008 (H.L.).

The general, but not conclusive, test for causation is the "but for" test, which requires the plaintiff to show that the injury would not have occurred but for the negligence of the defendant: *Horsley v. MacLaren*, [1972] S.C.R. 441.

The "but for" test is unworkable in some circumstances, so the courts have recognized that causation is established where the defendant's negligence "materially contributed" to the occurrence of the injury: *Myers v. Peel County Board of Education*; [1981] 2 S.C.R. 21, *Bonnington Castings, Ltd. v. Wardlaw*, [1956] 1 All E.R. 615 (H.L.); *McGhee v. National Coal Board*, *supra*....

[107] In the context of a delayed medical diagnosis, in *Cottrell v. Gerrard*, 2003 Can LII 50091 (ONCA) at para 25, the Ontario Court of Appeal described the test as follows:

...[I]n an action for delayed medical diagnosis and treatment, a plaintiff must prove on a balance of probabilities that the delay caused or contributed to the unfavourable outcome. In other words, if, on a balance of probabilities, the plaintiff fails to prove that the unfavourable outcome would have been avoided with prompt diagnosis and treatment, then the plaintiff's claim must fail. It is not sufficient to prove that adequate diagnosis and treatment would have afforded a chance of avoiding the unfavourable outcome unless that chance surpasses the threshold of "more likely than not".

[108] In this case, the Plaintiffs must establish that Dr. Kassim's failure to identify and report Mary Fleury's cancer more likely than not caused or contributed to her death.

[109] One of the circumstances in which the "but for" test is potentially unworkable is in medical malpractice cases where scientific proof of causation is simply not attainable. As such, courts are instructed to take a "robust and pragmatic" approach to the fact finding component of causation. *Benhaim v. St. German*, 216 SCC 48 at para 54. *Snell v. Farrell* [1990] 2 S.C.R. 311 (SCC) at para.34-38), *Clements (Litigation Guardian of) v. Clements*, 2012 SCC 32 at paras 9- 10 and 38.

[110] A “robust and pragmatic” approach should be applied to the evidence called at trial, remaining mindful of the fact that the plaintiff must call sufficient evidence to meet the onus it bears of establishing causation on a balance of probabilities. *Aristorenas v. Comcare Health Services*, 2006 CarswellOnt 6155 (ONCA), at paras 56-57, leave to appeal to SCC refused 2007 CarswellOnt 1878 SCC.

[111] An important aspect of the causation analysis in this case is the staging of Mary Fleury’s cancer in 2011. Unfortunately the best evidence as to the staging of Mary Fleury’s cancer is unavailable because her cancer was not diagnosed in 2011. Had Ms. Fleury been diagnosed with cancer in 2011, it is agreed by the experts, that further testing and treatment would have taken place. The evidence establishes that those further steps would have included:

- a. Her lymph nodes would have been removed and examined to determine if the cancer had spread to them;
- b. She would have had a right hemicolectomy and samples would have been taken and examined to see if the cancer had spread to the colon and tissue surrounding that area; and
- c. Samples would have been taken of her peritoneal cavity, including the fluid inside her abdomen. Those samples would have been examined to see if the cancer had spread into the peritoneal cavity.

[112] In cases of causal uncertainty, both parties face the difficulty of attempting to establish facts in the absence of complete information. In some circumstances, where the defendant’s negligence is the cause of the uncertainty, an adverse inference of causation may be drawn against the defendant. Whether an inference of causation is warranted, and how it is to be weighed against the evidence, are matters for the trier of fact. However, a court must be cautious in this approach in order to avoid creating a consequential reversal of the burden of proof. *Benhaim v. St. German*, 2016 SCC 48 at para 42 and 66-67.

[113] Given the totality of evidence in this case, I do not draw an adverse inference of causation. Both the Plaintiffs and Defendant are in the same position with respect to the unavailable evidence. The ability of the experts for both the Plaintiffs and the Defendant to opine regarding causation was impacted in the same way by the absence of the evidence. Both experts were ultimately able to offer opinions and conclusions based on the evidence that was available.

[114] The court heard from two expert witnesses on the issue of causation, Dr. Sharlene Gill and Dr. Mark David Vincent. Both agreed that the issue of causation in this case rests upon the probable stage of Mary’s cancer in 2011 and whether any of the offered treatments would have probably resulted in a cure.

A. Dr. Sharlene Gill

[115] The Plaintiffs called Dr. Sharlene Gill on the issue of causation. Dr. Gill is an oncologist who specializes in gastrointestinal cancer, which includes appendiceal cancer. Dr. Gill was qualified as an expert to provide opinion evidence with respect to the field of oncology, including opinion evidence on the causative effect of a delay in diagnosis of Mary Fleury's appendiceal cancer.

Qualifications

[116] Dr. Gill has over 20 years of experience as an oncologist and has been practicing with the BC Cancer Agency for most of that time. The majority of her work has centered around gastrointestinal cancer.

[117] Provincially, Dr. Gill is a member of the GI cancer outcomes unit in British Columbia, which is a research effort to use real world patient data to develop outcomes and provide evidence where randomized trial evidence is unavailable. She was the chair of the provincial GI Tumour group between 2007 and 2014; and co-chair of the provincial GI Cancers Outcomes Unit from 2016 to 2021.

[118] Nationally, Dr. Gill has been a member of the Canadian Cancer Clinical Trials Group, which is the national group for cancer trials in Canada. She is the co-chair of the NCI Intergroup Colon Cancer Task Force, on the executive of the Commonwealth Neuroendocrine Tumours, is the Chair of the CCTG Gastrointestinal Disease Site committee and is on the CCTG National Cancer Institute GI Steering Committee.

[119] Internationally, Dr. Gill is a member of the American Society of Clinical Oncology (ASCO) and the European Society of Medical Oncology (ESMO) and has sat on various committees related to GI cancers with both societies.

[120] Dr. Gill is currently a full-time professor in oncology at UBC, and an associate professor in the GI division. She has authored over 100 articles and book chapters on the subject of GI cancer. In 2009, she co-authored a paper specifically on the treatment of cancers of appendiceal origin in a highly referenced surgery journal. She is the editor in chief for *Current Oncology*, and an associate editor of *Therapeutic Advances in Medical Oncology*.

[121] Dr. Gill has treated patients with appendiceal cancer, including patients with goblet cell carcinoma (GCC) and adenocarcinoma ex GCC (AEGCC), which is the type of cancer that Mary Fleury had.

[122] Dr. Gill testified that she understood that it was her role to provide fair, objective and non-partisan evidence in her field of expertise and that this duty prevailed over any obligation she might feel towards the Plaintiffs.

Dr. Gill's ultimate opinion on causation

- [123] It was Dr. Gill's opinion that in October, 2011, Mary Fleury likely had stage IIB cancer of the appendix, meaning that it was localized with no evidence of metastatic spread.
- [124] It was Dr. Gill's evidence that in October, 2011, the standard treatment for Mary Fleury's type of cancer would have been surgery, in the form of a right hemicolectomy, followed by adjuvant chemotherapy.
- [125] It was Dr. Gill's opinion that, had Mary Fleury's cancer been diagnosed in October, 2011 and had she received the standard of care treatment, she would likely be alive today.
- [126] Dr. Gill explained that the type of cancer Mary had is called "adenocarcinoma ex goblet cell carcinoma" (hereinafter referred to as AEGCC). This is a mixed histology type of goblet cell cancer where there is a component of adenocarcinoma. It is a rare form of cancer.
- [127] Dr. Gill testified that for rare cancers, such as the one Mary had, oncologists don't have the benefit of large, randomized, and controlled trials with respect to prognosis and treatment. As such, oncologists must rely on the best available evidence.

Dr. Gill's opinion that Mary Fleury had Stage IIB cancer:

- [128] Dr. Gill testified that the most commonly recognized classification system oncologists use for cancer is the American Joint Committee on Cancer Staging system (AJCC). This is an internationally accepted system used for cancer prognosis.
- [129] All experts called at trial agreed that the AJCC Staging Manual (8th edition) is currently an authoritative manual used by oncologists when staging cancer. Chapter 19 of the AJCC staging manual (8th edition) pertains to the staging of carcinomas of the appendix, including high-grade neuroendocrine carcinomas, mixed adenoneuroendocrine carcinomas and goblet cell carcinoids. Chapter 19 was marked as Exhibit 13 at trial.
- [130] An important part of the AJCC staging system is the use of the TNM categories to record information about the presenting cancer. The TNM categories are as follows:
- T** - refers to the extent and size of the primary tumour;
 - N** - refers the extent of nodal involvement; and
 - M** – refers to whether there is any clinical evidence of distant metastases, or metastatic disease.
- [131] The T and N classification are determined by a pathologist who examines the tumour and lymph nodes. The M classification is based on clinical evidence.

[132] With respect to the **T** (tumour) classification, a pathologist examines the extent and size of the primary tumour. In this case the pathologists agreed that Mary Fleury's tumour would be classified as **T4a**, which means the tumour had extended to the serosal surface of the appendix. Using the analogy that an appendix is like a hose, this type of tumour is one that has reached the outside layer of the hose.

[133] A pathologist will also report on the tumour's histological grade. Histological grade refers to how abnormal the cells appear under the microscope. The more abnormal the cells look, the higher the grade. A high-grade tumour is also referred to as "poorly differentiated". Dr. Gill testified that what this means is that the cancer is more aggressive, and in patients with Stage IV cancer it is a poor prognostic factor. In Mary's case, her tumour was classified as G3, or "poorly differentiated".

[134] With respect to the **N** (nodal) classification, a pathologist will examine the lymph nodes and report the extent to which the cancer has involved the lymph nodes, and how many are involved. Dr. Gill described the lymph nodes as a "highway" through which cancer can spread.

[135] With respect to the **M** (distant metastases) classification, Dr. Gill testified that this is determined by the existence of clinical evidence of distant metastatic spread to other parts of the body.

[136] Using the AJCC classification system, Dr. Gill testified that she would classify Mary Fleury's cancer in 2011 as PT4a (type of tumour), N0 (no nodal involvement) and M0 (no distant metastasis). Using the AJCC Staging Manual (8th edition), a patient with **PT4aN0M0** appendiceal cancer would be diagnosed as having Stage IIB cancer.

[137] Dr. Gill testified that she relied on the pathologists' opinions with respect to the T classification. At trial all the pathologists agreed it would have been classified as PT4a.

[138] Dr. Gill testified that she relied on the following facts to support her opinion that Mary likely had no nodal involvement in 2011 and would have been staged as **N0**:

1. There was no clinical evidence from 2011 that the lymph nodes were involved;
2. When Dr. Kassim examined Mary Fleury's lymph nodes in 2015 he found no evidence of nodal involvement; and
3. Given the progression of the disease, it was highly unlikely that there was lymph node involvement in 2011.

[139] Dr. Gill testified that she relied on the following facts to support her opinion that Mary likely had no distant metastatic spread in 2011 and would have been staged as **M0**:

1. Mary's cancer in 2011, was an incidental finding. Mary had a fecal blockage in her appendix, which caused her to develop appendicitis. She did not present with clinical evidence of appendiceal cancer;
2. The CT scan conducted in October, 2011 showed no clinical evidence of metastatic spread;
3. The CT scan conducted in October, 2011, showed no evidence of ascites. Ascites are fluid that typically builds up in the abdomen if there has been widespread peritoneal spread;
4. In December, 2011, Mary underwent a total abdominal hysterectomy and bilateral oophorectomy, which meant her uterus, fallopian tubes and ovaries were removed. Those organs were examined and there was no evidence of cancer;
5. Where there is peritoneal spread, the most common area the cancer will spread to is the nearby ovaries. Mary Fleury had no evidence of cancer in her ovaries in 2011;
6. In December, 2011, the surgeon performed a pelvic washing. A pelvic washing consists of irrigating the inner lining of the abdomen and then aspirating the fluid. The fluid is then examined for the presence of abnormal or cancerous cells. In this case, there was no evidence of cancer detected in the fluid from the pelvic washing;
7. When an AEGCC, grade 3 tumour metastasizes it spreads very quickly and one would expect the presentation of symptoms. Mary did not have any symptoms of metastatic spread in 2011;
8. Given the aggressive nature of this type of cancer, it was "nearly impossible" that Mary Fleury had untreated metastatic disease for four years, without any clinical symptoms or diagnosis of cancer; and
9. It would be "unreasonable to expect" someone who had an aggressive form of stage IV AEGCC cancer in 2011 to live five years, without any treatment. This would be significantly outside the recognized prognosis for someone with stage IV AEGCC cancer.

[140] Dr. Gill testified that it is fundamental in cancer that the earlier you pick it up the better because early detection impacts prognosis. Dr. Gill testified that had Mary Fleury received the definitive treatment for Stage IIB AEGCC, she would be alive today.

[141] In cross-examination, Dr. Gill agreed that a T4a tumour meant that the cancer had spread to the outside lining of the appendix. She agreed that this meant that the cancer was exposed to the inside of the abdomen and such it was “possible” that the cancer had established metastases in 2011.

[142] Dr. Gill acknowledged in cross-examination that she was aware of Dr. Ali-Ridha’s opinion that there was lymph node involvement in 2015. However, she testified that this did not materially impact her opinion with respect to the likelihood of nodal involvement in 2011.

[143] In cross-examination, Dr. Gill testified that it was her opinion that metastasis had likely occurred within a few months of Mary’s diagnosis in 2015. She testified that given this was a poorly differentiated tumour, once metastasis sets in you would expect the cancer to spread quickly. She agreed with the suggestion that metastasis could have begun as early as 6-12 months prior to diagnosis.

[144] It was suggested in cross examination, and Dr. Gill agreed, that because Mary Fleury presented with stage IV cancer in 2015, cancer had to have been left behind when her appendix was removed. It was agreed that three scenarios must have occurred in 2011:

1. The cancer had spread directly from the appendix to other organs;
2. The cancer had spread to the nodes, and then spread further from there; or
3. The cancer had spread microscopically into the peritoneal cavity, typically starting in the area of the appendix and then continued to spread from there.

[145] Dr. Gill agreed with the suggestion that it was unlikely that either the first or second scenario had occurred given that the 2011 proximal margin did not show evidence of cancer which suggests that there had not likely been direct spread to another organ. Further, there was no evidence of nodal involvement in 2011.

[146] With respect to the third scenario, Dr. Gill agreed that it is common that this type of cancer will spread into the abdominal cavity, typically starting in the area of the appendix. It is for this reason that oncologists recommend a right hemicolectomy followed by chemotherapy. A right hemicolectomy is expected to remove the microscopic residual cancer in the area of the appendix. Adjuvant chemotherapy then kills any remaining, microscopic cancer cells.

[147] Dr. Gill agreed that had the lymph nodes been removed in 2011, and they were pathologically confirmed to be positive, that Mary Fleury would have had stage III cancer.

- [148] Dr. Gill agreed during cross examination that there were features of Mary Fleury's tumour that made it a higher risk for metastases. The tumour was greater than or equal to 2 centimetres, it had a positive margin, and it presented as PT4a which meant it spread to the outside lining of the appendix.
- [149] Dr. Gill testified that not everyone with AEGCC has metastatic cancer. Further, that the fact that Mary's tumour was a PT4a did not mean it was a foregone conclusion that there was widespread peritoneal involvement. It meant that there was an increased risk of microscopic metastatic disease.
- [150] Dr. Gill was cross-examined with respect to the *Tang* study, which was referenced by Dr. Vincent in his report and testimony. This study was published in 2008 and was based on a retrospective study where researchers reviewed the medical files of 63 patients who had presented with goblet cell carcinoma at the Sloan-Kettering Cancer Centre in New York. This article was marked as Exhibit 17, *Tang et al. Pathologic Classification and Clinical Behavior of the Spectrum of Goblet Cell Carcinoid Tumours of the Appendix, AM J Surg Pathol, Volume 32, Number 10, October, 2008*.
- [151] In this study, the researchers divided the GCC patients into three classifications, Tang A, Tang B and Tang C, depending on the pathological characteristics present in the cancer. Patients with AEGCC were placed into the "Tang C" category. Dr. Gill agreed that Mary Fleury's cancer would likely be classified as a "Tang C".
- [152] It was suggested to Dr. Gill in cross examination that because all of the "Tang C" patients in this study died prior to the five-year mark, that Mary Fleury's prognosis would have been very poor in 2011. Dr. Gill disagreed with this suggestion.
- [153] Dr. Gill testified that that the *Tang* study was a non-randomized retrospective study where the authors simply looked at the medical files of 63 patients as they presented at the hospital. Out of the total number of patients in the study, only seven were classified as "Tang C" and all presented with stage IV disease. The study did not have any "Tang C" patients who presented with stage I, II or III cancer.
- [154] Dr. Gill testified that one could not infer that because the seven "Tang C" stage IV cancer patients died, it was more likely that a patient with "Tang C" stage IIB cancer would likely die.
- [155] Keeping in mind the limitations of the *Tang* study, Dr. Gill pointed out that the mean survival rate for the "Tang C" stage IV patients in the *Tang* study was 31 months, with all patients having died within 5 years. Mary Fleury lived 62 months, untreated. This was well beyond the mean survival rate for "Tang C" stage IV patients and was one of the reasons Dr. Gill concluded that it was unlikely that Mary Fleury had stage IV cancer in 2011.
- [156] Dr. Gill was cross-examined with respect to the *Reid* study, which was also referenced and relied upon in Dr. Vincent's report and testimony. The *Reid* study was published in 2015 and was based on a retrospective study where researchers

reviewed the medical files of 77 patients who had AEGCC. This article was marked as Exhibit 18, *Reid et.al Mod Pathol. 2016 October; 29(10);1243-1253*

[157] It was suggested in cross-examination that this study supported an inference that Mary Fleury likely had a poor prognosis in 2011 because the median survival for the patients reviewed in that study was only 38 months. Dr. Gill disagreed with this suggestion.

[158] Dr. Gill testified that the paper did not provide a stage specific survival rate. In the study, 65% of the patients had stage IV cancer, and 7% had stage III cancer. Given 72% of the patients had a more advanced stage of cancer than Mary Fleury, it could not support an inference that Mary Fleury would have had a similar prognosis.

[159] Dr. Gill further pointed out that the median range of survival in that study was 38 months. Given that the majority of patients reviewed in the *Reid* study had stage IV cancer, this was another factor that made it “very, very unlikely” that Mary Fleury had stage IV cancer in 2011, because she went on to live for 62 months, untreated.

Dr. Gill’s opinion on the effectiveness of treatment

[160] Dr. Gill testified that in oncology, experts talk about the metric of five-year survival, because traditionally people alive and free of disease five years from the time of diagnosis are considered cured. It was her opinion that had Mary Fleury received the definitive treatment her five-year risk of recurrence would have been less than 30%.

[161] It was Dr. Gill’s opinion that had Mary Fleury received the standard of care treatment in 2011, given the probable stage of her cancer, she would likely be alive today.

[162] Dr. Gill testified that because appendiceal cancer is rare, and there are no randomized and controlled trials, oncologists rely on the best available evidence when recommending treatment.

[163] The “best available evidence” relied on by oncologists includes their understanding of the biology of the disease and reasonable scientific extrapolation to other similar disease settings.

[164] Both Dr. Gill and Dr. Vincent agreed that the definitive recommended treatment in 2011 would have been to undergo a right hemicolectomy followed by adjuvant chemotherapy. Dr. Gill testified that for GI cancers, particularly intestinal cancers, the recommended chemotherapy would have involved the use of a medication called FOLFOX.

[165] Dr. Gill testified that the goal of the right hemicolectomy is to remove the right half of the colon as well as the lymph nodes. The intent of this surgery is to remove any areas, including the mesentery (the connective tissue surrounding the appendix and

colon) where residual cancer cells would hide. The lymph nodes are also removed to address the risk of microscopic disease.

[166] Dr. Gill testified that the goal of adjuvant chemotherapy is to target any remaining cancer cells and kill them. It is used to eradicate any microscopic presence of the disease left over after the hemicolectomy and to prevent recurrence in the future.

[167] Dr. Gill testified that a right hemicolectomy is widely endorsed in the guidelines of the North American Neuroendocrine Tumour Society, the European Neuroendocrine Tumour Society and the American Society of Clinical Oncology.

[168] Dr. Gill testified that adjuvant chemotherapy is recommended based on established benefits proven in large, randomized trials for colon cancers and intestinal cancers. She testified that Goblet Cell Carcinomas (GCC's) fall somewhere in between adenocarcinoma and neuroendocrine carcinomas. Experts extrapolate benefits based on the benefits of adjuvant chemotherapy for intestinal adenocarcinomas. Dr. Gill testified that this is not a far-fetched extrapolation, rather it is very closely linked.

[169] Dr. Gill testified that she strongly disagreed with Dr. Vincent's opinion that the definitive recommended treatment would provide no significant benefit to a cancer patient with AEGCC. It was her evidence that this was inconsistent with multiple international guidelines.

[170] Dr. Gill testified that it would not be reasonable to suggest that oncologists would offer the standard of care treatment which involves undergoing invasive surgery and toxic chemotherapy, with the knowledge that it would have no significant benefit.

[171] In cross-examination, counsel for the Defendant pointed out that Dr. Gill had not referenced any studies that showed the benefit of the recommended treatment for AEGCC. Dr. Gill acknowledged that there is a paucity of data in this area. As such, oncologists must rely on the best available evidence, which includes looking at the totality of the evidence and using scientific extrapolation.

[172] Dr. Gill acknowledged that in some circumstances cancer treatment has been offered to a patient that might not likely result in a cure.

[173] Dr. Gill was cross-examined on the *Pham* study, which was relied on by Dr. Vincent in his opinion. The *Pham* study was published in 2006 and consisted of a retrospective study of patients who underwent treatment for GCC at the Mayo Clinic. A copy of this study was marked as Exhibit 20, *Pham et. al. Annals of Surgical Oncology, 13(3): 370-376*.

[174] It was suggested in cross-examination that the *Pham* study demonstrated that a hemicolectomy has little impact on survival for patients with GCC. Dr. Gill disagreed and testified that the *Pham* study must be viewed with caution because this was a non-random retrospective study of patients, some of whom had only had an

appendectomy. Further, any reliance on this study would be impacted by the prognostic factors that existed for each patient that had a hemicolectomy. Ultimately the researchers in that study concluded that they were not able to calculate the probability of recurrence for each stage of cancer based on those who had an appendectomy versus right hemicolectomy.

- [175] With respect to the non-randomized retrospective studies relied on by Dr. Vincent in support of his opinion on the effectiveness of treatment, Dr. Gill cautioned that you can't infer that there is no benefit from treatment based on these studies. You can only say that in that particular study, with those particular patients, they did not see a difference.

B. Dr. Mark David Vincent

- [176] On the issue of causation, the Defendant called Dr. Mark David Vincent. Dr. Vincent is an oncologist who specializes in thoracic cancer and gastrointestinal cancer. Dr. Vincent was qualified as an expert to provide opinion evidence with respect to the field of oncology, including opinion evidence on the causative effect of any delay in diagnosis of Mary Fleury's appendiceal cancer.

Qualifications

- [177] Dr. Vincent has over 30 years experience as an oncologist and has spent his career at the London Health Science Centre in London, Ontario. He has been part of the thoracic and gastro-intestinal multidisciplinary team at that hospital since 1991. He is currently the thoracic disease site team chair with the London Regional Cancer program. He sits on various advisory committees for a number of private companies.
- [178] Dr. Vincent is a member of the American Society of Clinical Oncology, and the European Society of Medical Oncology. Between 1993 and 2000, he was a member of the phase I working group, with the National Cancer institute of Canada.
- [179] Dr. Vincent is currently a professor with the department of oncology at the Schulich School of Medicine and Dentistry at Western University and has published and presented primarily on thoracic cancer. He has not published on appendiceal cancer because it is a very rare form of cancer. He is currently on the editorial board for *Current Oncology* and is a journal reviewer for other publications.
- [180] Dr. Vincent has dealt with patients who have appendiceal cancer and is currently treating three patients with goblet cell cancer.
- [181] Dr. Vincent testified that he understood that it was his role to provide fair, objective and non-partisan evidence in his field of expertise and that this duty prevailed over any other obligations in this matter.

Dr. Vincent's ultimate opinion on causation

[182] Dr. Vincent testified that it was his opinion that in October, 2011, Mary Fleury likely had stage IV cancer of the appendix. It was his opinion that the cancer had likely already microscopically spread widely throughout her peritoneal cavity before her appendix was removed.

[183] It was Dr. Vincent's opinion that Mary Fleury would not be alive today, regardless of whether she had been diagnosed with and treated for cancer in 2011.

Dr. Vincent's opinion that Mary Fleury had Stage IV cancer

[184] Dr. Vincent testified that he considered the following factors to be important in his opinion that Mary Fleury likely had stage IV cancer:

1. The 2011 tumour was staged pathologically as PT4a. This meant that the cancer had grown through the wall of the appendix and was exposed to the peritoneal cavity. Once exposed to the peritoneal cavity, there was no barrier to the cancer spreading widely;
2. Dr. Kassim classified the 2011 tumour in his amended report as "Pt4aNxM1a", which meant he had pathologically determined that there had been metastatic spread in 2011;
3. In 2011 Mary Fleury's appendix had ruptured, and this was another method for the cancer to spread into her peritoneal cavity;
4. In 2015, it was determined that Mary Fleury's cancer had spread into her bowels from the outside, this meant that the cancer had spread through her peritoneal cavity and landed on the bowels;
5. The 2011 tumour was poorly differentiated;
6. The 2011 tumour had a high mitotic index;
7. Mary Fleury's gender (female) was a poor prognostic factor, citing the *Reid* study;
8. Mary Fleury was under the age of 55, which was a poor prognostic factor, citing the *Reid* study; and
9. Mary Fleury had either a Tang B or Tang C tumour. All of the patients classified as "Tang C" in the *Tang* study had stage IV

metastatic disease. As such, it was more likely that Mary had stage IV cancer; and

10. Peritoneal spread is very common in GI cancers and when it happens it is usually incurable.

[185] It was Dr. Vincent's opinion that it was more likely that Mary Fleury's cancer would be staged as **PT4aNxM1a** which would be classified using the AJCC Staging Manual as stage IV appendiceal cancer.

[186] Dr. Vincent's opinion was that it was more likely that microscopic cancer cells had spread widely throughout the abdominal cavity in 2011. He testified that because the abdomen can accommodate many tiny, diffuse tumours, this could mean that you would not see symptoms for some time. He testified that whether cancer was seen or not in 2011, was just a technicality.

[187] Dr. Vincent differentiated his opinion from Dr. Gill on the basis that while there was no "overt metastatic spread", he felt that there was unseen "microscopic spread" which in his opinion would classify as distant metastatic spread (the M, in the TNM staging process).

[188] During cross-examination it became clear that Dr. Vincent had misunderstood Dr. Kassim's amended report and evidence with respect to the staging of the tumour in 2011. The evidence at trial was that Dr. Kassim had classified the **2015** cancer as "Pt4aNxM1a". There was no issue that Mary Fleury had stage IV cancer in 2015.

[189] Dr. Kassim testified at trial that he found no evidence of distant metastatic spread when he re-examined the **2011** slides. This was brought to Dr. Vincent's attention during cross-examination.

[190] With respect to Dr. Vincent's reliance on the fact that Mary Fleury had a burst appendix in 2011 to support his conclusion that she had stage IV cancer, it was pointed out in cross-examination that the literature that Dr. Vincent relied on suggested that patients who present with appendicitis are more frequently diagnosed with stage I or stage II cancer. Further, the literature suggested that there are lower rates of peritoneal metastatic spread where appendicitis was present.

[191] When asked why Dr. Vincent did not bring this to the court's attention, Dr. Vincent provided various answers which included that he thought he had put this in his second report, however when he looked at his second report it was not there; that he thought Dr. Gill had put this in her report; and that ultimately, he didn't believe it was highly relevant.

[192] During cross-examination Dr. Vincent agreed that the most common area where peritoneal spread will manifest itself is in the ovaries. When counsel for the Plaintiffs pointed out that in 2011 there was no evidence of cancer in Mary Fleury's ovaries, Dr. Vincent seemed to suggest that he was not prepared to rely on the original 2011

pathology report from Soldier's Memorial Hospital or Dr. Kassim's re-examination of the ovaries in 2015 as being necessarily accurate. Dr. Vincent cited his lack of knowledge as to the pathologists' expertise as a basis to not necessarily accept their findings.

[193] Dr. Vincent testified that while he relied on the fact that Mary's cancer had a high mitotic index in support of his opinion, he would defer to the opinion of Dr. Lage on this issue. Dr. Lage testified that pathologists no longer use the mitotic index as a prognostic factor. Instead, they rely on the finding that the tumour was "poorly differentiated" which is a poor prognostic factor. This was consistent with the evidence of Dr. Gill.

[194] Dr. Vincent testified that he relied on the *Reid* study in support of his opinion that Ms. Fleury's gender was a poor prognostic factor. In cross-examination it was pointed out that the authors of the *Reid* study found that gender difference was no longer statistically significant after controlling for other variables. Despite this, Dr. Vincent insisted it was still a significant factor to him.

[195] Dr. Vincent was cross-examined extensively on his reliance on the *Tang* study to support his opinion that because the Tang C patients in that study had stage IV cancer, it was more likely that Mary Fleury had stage IV cancer. When counsel for the Plaintiffs attempted to point out aspects of the *Tang* study that were favourable to the Plaintiffs' case, such as the fact that most of the Tang C stage IV patients were dead within three years, Dr. Vincent appeared to distance himself from the article stating "You're making it an extrapolation from a small seven group of – seven patients to an individual, answer is on the basis of seven patients. I don't think that doctors generally make those kind of generalizations".

Dr. Vincent's opinion with respect to treatment

[196] Dr. Vincent agreed that the standard treatment that would have been offered to Mary Fleury in 2011 would have been surgery (a right hemicolectomy) and adjuvant chemotherapy.

[197] Dr. Vincent testified that despite this being the standard accepted treatment, it was his opinion that this "so-called" treatment would have had no substantive impact on the outcome in this case and Mary Fleury likely would have died in any event.

[198] Dr. Vincent testified that he would not have advised Mary Fleury to undergo this treatment. He would tell a patient in her position that it might possibly work, but probably wouldn't work and that he did not think it would. Ultimately it would be the patient's decision, balancing the risks involved. Many patients will undergo treatment, even if there is only a small chance of benefit.

[199] According to Dr. Vincent, a right hemicolectomy and adjuvant chemotherapy would have no substantive impact on individuals who have appendiceal cancer. In support of his opinion, he relied heavily on what he referred to as his review of "the literature".

This consisted of published articles related to retrospective reviews of patient information from various hospitals.

[200] Dr. Vincent testified that he relied upon the following findings in the literature to support his opinion:

1. In the “*Lamarca study*” the authors found that none of the patients they reviewed benefitted from a hemicolectomy. In fact, the patients tended to do worse. Exhibit 37 - *Lamarca et al, 2015 Appendiceal Goblet Cell Carcinoids: Management Considerations from a Reference Peritoneal Tumour Service Centre and ENETS Centre of Excellence. Neuroendocrinology 2016; 103:500-517.*
2. In the “*Lamarca study*” the authors found that those who had adjuvant chemotherapy had a shorter period of disease-free survival.
3. In the “*Pham study*” the authors found that a hemicolectomy or chemotherapy did not make a difference with respect to survival for patients with stages II, III, and IV GCC cancer. Exhibit 20 - *Tuan Pham et. al, 2006, Annals of Surgical Oncology 13 (3); 370-376.*
4. In the “*Jessinghaus study*” the authors found that goblet cell carcinoids and adenocarcinomas ex goblet are genetically distinct from primary colorectal type adenocarcinoma. As such, you cannot draw a link between the effectiveness of chemotherapy for colon cancer to its effectiveness for appendiceal cancer. Exhibit 38 *Jesinghaus et al, 2018, Appendiceal Goblet Cell Carcinoids and Adenocarcinomas Ex-goblet cell Carcinoid are Genetically Distinct from primary colorectal-type adenocarcinoma of the appendix. Modern Pathology 2018 31, 829-839.*

[201] Dr. Vincent was dismissive of Dr. Gill’s evidence with respect to use of FOLFOX adjuvant chemotherapy for patients with appendiceal cancer given its proven effectiveness in large, randomized trials for colon and intestinal cancer. Dr. Vincent testified that the “thinking, if you would call it that” that because the appendix is attached to the colon and so the responsiveness would be the same as in the case of colon cancer was not supportable. Dr. Vincent testified that there is now “recent and good evidence” that the molecular pathology or at least the way that GCC’s become cancerous is quite different than the way colon cancers become cancerous. He then referred to the *Jessinghaus study* in support of this.

[202] With respect to the *Jessinghaus* study, counsel for the Plaintiffs pointed out that the authors of that study noted in their article that their findings were at variance with a similar prior genetic study. The authors stated that they did not have a ready explanation for the discrepancy, however presented a possible a theory about why their study came to a different conclusion.

[203] Dr. Vincent agreed that he did not bring the contrary opinion to the court's attention in his prior testimony. His explanation was that he missed this statement in the article.

[204] In cross-examination, Dr. Vincent agreed that the "*Pham*" study would now be considered out of date because the types of treatments that would have been provided between 1984 and 2004 would now be outdated. In particular the type of chemotherapy used had changed by 2011.

C. Analysis on Causation:

[205] Before turning to the analysis with respect to the staging of Mary Fleury's cancer and the effectiveness of treatment, the court makes the following findings with respect to the evidence of Dr. Gill and Dr. Vincent.

[206] The court finds that the evidence of Dr. Gill was clear, convincing and supported by the evidence available in this case. Dr. Gill conceded, where appropriate, points raised by counsel for the Defendant. She testified in a responsive manner. Having listened to and observed Dr. Gill's testimony, the court was satisfied that she was providing her evidence in accordance with her duty to be fair, objective and non-partisan in her field of expertise.

[207] The court finds that Dr. Vincent's evidence was not always objective or balanced. During cross-examination it became apparent that he did not feel that he had to refer to aspects of the literature that supported aspects of the Plaintiffs' case. When it was pointed out that some of the studies he relied on supported aspects of the Plaintiffs' case he would distance himself from the study and explain that it was a non-randomized retrospective study from which one could not make generalizations.

[208] Dr. Vincent was, at times, argumentative during his cross-examination, posing questions back to counsel, interrupting counsel, or directing counsel to "get to the point". At one point during the trial the court was required to take a break in the proceedings to defuse the argumentative nature of Dr. Vincent's evidence.

[209] Overall, the court had significant concerns about whether Dr. Vincent understood his duty to be non-partisan and to provide fair and balanced evidence on the issues.

Staging:

[210] Having listened to the evidence of Dr. Gill and Dr. Vincent, both in its content and the manner of presentation, the court accepts Dr. Gill's evidence with respect to the staging of Mary Fleury's cancer in 2011.

[211] Dr. Gill's opinion that Mary Fleury likely had Stage IIB cancer is supported by objective facts and observations made between 2011 and 2015. Those objective facts included:

1. Dr. Kassim found no evidence of distant metastasis when he examined the 2011 appendiceal samples, both in 2011 and 2015;
2. Dr. Ali-Ridha found no evidence of distant metastasis when he examined the 2011 appendiceal samples;
3. While Dr. Lage found evidence that the cancer may have spread to the surrounding appendiceal mesentery, this is within centimetres of the appendix. No one opined that this would amount to "distant" metastasis;
4. Mary Fleury underwent a hysterectomy and an oophorectomy in 2011, less than 3 months following the appendectomy. At that time surgeons removed Mary's uterus, her two ovaries and fallopian tubes. These organs are adjacent to the appendix, and are the most common area where appendiceal cancer will spread. The ovaries, fallopian tubes and uterus were examined by a pathologist in 2011 and they found no evidence of cancer;
5. The samples from Mary Fleury's hysterectomy and oophorectomy were re-reviewed in 2015, with knowledge that Mary Fleury now had appendiceal cancer. Again, no evidence of cancer was found;
6. During the hysterectomy, the surgeon took "washings" from the inside of Mary Fleury's peritoneal cavity. The peritoneal cavity is another very common area where appendiceal cancer will spread. When these "washings" were examined pathologically, they showed no evidence of cancer;
7. Mary Fleury exhibited no symptoms of metastatic spread from 2011 to 2015 (about 3 years). The court accepts Dr. Gill's evidence that this is inconsistent with someone experiencing the aggressive metastatic spread expected at Stage IV of this disease; and
8. Mary Fleury lived a further 62 months following her appendectomy. Living for this length of time with untreated stage IV cancer is inconsistent with every study referred to during this trial. While the court recognizes the limitations of the studies and that some patients live longer than expected, this is

another factor that the court considers in combination with the others listed above.

[212] Applying a robust and pragmatic approach to the evidence called at trial, the court concludes that it is unlikely that Mary Fleury had stage IV cancer in 2011.

[213] With respect to Dr. Vincent's theory with respect to "microscopic" spread, as opposed to "overt" spread, the court finds that it would be speculative, at best, to rely on this theory. Indeed, Dr. Vincent conceded in cross-examination that no one would be given a stage IV diagnosis based on the "possibility" of metastatic spread. The court accepts that a stage IV diagnosis is provided when there is distant metastasis observed either clinically or by a pathologist. In this case there was no observed clinical or pathological evidence of metastatic spread in 2011.

[214] It was concerning to the court that when Dr. Vincent suggested that a burst appendix supported a stage IV diagnosis, he would not also include in his testimony that studies show patients who present with appendicitis tend to present with earlier stages of cancer and have lower rates of peritoneal spread. His explanation that he didn't mention this to the court because he thought it was in Dr. Gill's report demonstrated that he saw himself more as an advocate in this trial.

[215] It was also concerning that Dr. Vincent appeared reluctant to accept the findings of the pathologists, who had been part of Mary Fleury's treatment between 2011 and 2015, when their findings did not support his opinion. An example of this was when Dr. Vincent was asked if he was prepared to accept the findings of the pathologists at Mount Sinai with respect to the amount of adenocarcinoma found in Mary's cancer. Despite not being a pathologist, nor there being any evidence to the contrary, he testified as follows: "That was one opinion, but I don't think it was representative of the truth".

Treatment

[216] With respect to whether the standard, definitive treatment would have likely cured Mary Fleury, the court accepts the evidence of Dr. Gill. Dr. Gill presents with a higher degree of expertise in the area of GI cancers, having specialized in this area for some time as evidenced in her curriculum vitae and her publications.

[217] Dr. Gill is a member of the GI cancer outcomes unit in British Columbia, which is a research effort to use real world patient data to develop outcomes and provide evidence where randomized trial evidence is unavailable. Given how rare appendiceal cancer is and the lack of randomized trials on the efficacy of treatment, her experience in this area was valuable to the court.

[218] Dr. Gill's evidence with respect to treatment and its effectiveness was fair, balanced and reflected the widely accepted standards in oncology related to the treatment of appendiceal cancer.

- [219] The court appreciates, and Dr. Gill acknowledged, that not all substantive benefits of treatment include a cure. It may be that a patient will chose to undergo the treatment to simply extend their life. However, the evidence of Dr. Gill was that the substantive benefit to Mary Fleury would have been a cure.
- [220] The court accepts Dr. Gill's evidence that had Mary Fleury received the definitive treatment for appendiceal cancer, her five-year risk of recurrence would have been less than 30%. While the court cannot conclude with absolute certainty that Mary Fleury would have been cured of cancer, the court does conclude that it is more probable than not that she would have been and would be alive today.
- [221] Dr. Gill acknowledged in her evidence that there is a paucity of randomized clinical studies of appendiceal cancer because this type of cancer is very rare. As such, oncologists must turn to the best available evidence which includes their understanding of the biology of the disease and reasonable scientific extrapolation to other similar disease settings.
- [222] Dr. Gill testified that a hemicolectomy and adjuvant chemotherapy are recommended for appendiceal cancer found incidentally following an appendectomy, and that these recommendations are found in the guidelines published by the European Neuroendocrine Society, the North American Neuroendocrine Society and the American College of Clinical Oncology.
- [223] The standard treatment in 2011 would have meant that Mary Fleury would have had a right hemicolectomy which would have included the removal of the mesentery around the colon which contains the lymph nodes in that area. The purpose of this surgery is to clear out all the draining mesentery and lymph nodes to ensure any residual cancer is removed. Mary Fleury would then have received chemotherapy, which involves the administration of a combination of drugs that target and kill cancer cells and prevents them from proliferating and growing. The goal is to eradicate any microscopic residual disease.
- [224] The standard treatment in 2011 would have addressed the first, second and third scenario put to Dr. Gill in cross-examination with respect to cancer being left behind following the appendectomy. Scenario one, direct spread, would be addressed by removing any cancer that had spread directly to the colon. Scenario two, lymph node involvement, would be addressed by removing the lymph nodes thus eliminating the highway through which cancer can spread. Scenario three, residual cancer in the area of the appendix, would be addressed by removing the colon and the mesentery and following up with chemotherapy to eradicate microscopic spread.
- [225] The court had difficulty with Dr. Vincent's evidence that the standard treatment currently offered by oncologists to patients with appendiceal cancer provides no substantive benefit and may actually do more harm than good.

- [226] There was no evidence called at trial that would suggest that Dr. Vincent's opinion is widely accepted amongst oncologists. Dr. Gill testified that his opinion is contrary to multiple international guidelines.
- [227] The court accepts Dr. Gill's evidence that it is not reasonable to suggest that a standard of care treatment, based on best available evidence, is being offered by oncologists with knowledge that it has no benefit. The court does not accept that oncologists worldwide are currently having patients undergo invasive surgery and chemotherapy with the knowledge that it will provide no substantive good and may in fact be causing harm.
- [228] Dr. Vincent's opinion that treatment would not have made a difference to Mary Fleury's survival was almost exclusively based on what he referred to as the "literature". It was on this basis that he testified that he, personally, would not recommend the definitive, recommended treatment.
- [229] Dr. Vincent testified that the gold standard of academic investigation is randomized clinical trials which have enough patients to have adequate statistical power to conclusively draw relationships between cause and effect in terms of benefit or lack of benefit. Dr. Gill agreed with this and cautioned against drawing generalizations from non-randomized studies.
- [230] Despite Dr. Vincent's testimony with respect to randomized trials having adequate statistical power to conclusively draw relationships, he went on to say that it was his opinion was that "on average, looking at the aggregate of the data, it does not appear **statistically** that it would likely have exerted a major substantive effect". He then went on to refer to the "literature" in support of his opinion, citing primarily the non-randomized studies of *Lamarca*, *Tang*, *Pham*, and *Jessinghaus*.
- [231] Quite apart from the limitations of these studies pointed out in cross-examination, any "statistic" reliance on these studies must be regarded with caution given they do not have adequate statistical power to conclusively draw relationships between benefit or lack of benefit.
- [232] On the issue of treatment, Dr. Vincent's evidence continued to suffer from an inclination to only refer to information that would support the Defendant's case. For example, it was concerning that Dr. Vincent would testify that there is "now recent and good evidence" that the way that GCC's become cancerous is quite different than the way colon cancers become cancerous without bringing to the court's attention that in the study he relied on the authors acknowledged that its findings were contradicted by another study. Dr. Vincent's explanation that he "missed that part" was not reassuring. It would have been much more balanced and unbiased to acknowledge that there are two different studies that reach different conclusions and then explain why he preferred one over the other.

[233] The court is satisfied, having considered the totality of evidence called at trial, that the Plaintiffs have established, on a balance of probabilities, that but for Dr. Kassim's failure to meet the standard of care expected of an anatomic pathologist in similar circumstances, Mary Fleury would likely have sought out treatment for cancer and would be alive today.

VII. DAMAGES:

[234] Given that this court has concluded that Dr. Kassim did not meet the standard of care when he failed to diagnose Mary Fleury's cancer in 2011, and has concluded that but for this failure, Mary Fleury would be alive today, the Plaintiffs are entitled to damages.

[235] The Plaintiffs called four fact witnesses with respect to the issue of damages: George (Blair) McAmmond, Darlene Fleury, Patricia Sideris and David Brunatti. The court found all four witnesses to be both credible and reliable in the giving of their evidence. None of the witnesses were seriously challenged in cross-examination.

[236] The bulk of evidence that related to the issue of damages came from George McAmmond. The court found George McAmmond to be honest and forthright in his evidence. His evidence was fair and balanced. He conceded almost every point put to him in cross-examination. The court found him to be a credible and reliable witness.

[237] For the purposes of assessing damages, the court makes the following findings of fact based on the evidence it heard and accepts.

[238] Mary Fleury and George McAmmond re-connected in the summer prior to 2000, at a family event. Three months later they moved in together in Bowmanville. Mary brought her three children, Kathleen, Jordan and Megan. George brought his son, Kevin. Together, they raised their four children as a blended family.

[239] Around 2004 or 2005, the family decided to move to Parry Sound to be closer to family and to enjoy the type of lifestyle that a smaller, northern community could offer.

[240] George was able to secure employment with road construction right away. Mary initially worked in telecommunication, then cleaning before finally landing an office position with a local business, Brunatti Brothers. Mary worked for the Brunatti brothers for approximately 11 years, until she became too ill with cancer to continue working.

[241] One of Mary's employers, David Brunatti, testified that Mary was a well respected and appreciated member of their family business. He described Mary as "his left and right arm". Mary was an excellent employee and had a job at the company for as long as she wanted.

- [242] George's job was seasonal, working 6-12 months a year, depending on the work that was available. When they first moved to Parry Sound, George often worked out of town. As he developed seniority in his position, the amount of out-of-town work decreased.
- [243] When George worked out of town, Mary took care of the children and the home during the week. George would return on weekends and would assist with cooking and did his own laundry. George did most of the outdoor handyman work and maintenance. George testified that when he was in town the housework was split about 60:40, with Mary doing the larger part.
- [244] George testified that in terms of family expenses, he and Mary would pool the money they earned and would split the bills 50/50.
- [245] Just as Mary and George were starting to get ready to become "empty nesters" with their last child transitioning out of the house, Mary became aware that one of her daughters was unable to care for her two children, Damian and Gavin. Neither child had a father figure in their lives.
- [246] Damian began living with Mary on a permanent basis after her daughter agreed to give full custody to Mary. After a few years, Mary became aware that her second grandchild, Gavin, was in foster care. Mary contacted child welfare authorities and arranged to have full custody of Gavin as well. George and Mary took on the role of full-time parents for the two young boys.
- [247] Gavin is a special needs child. He was born with both physical and mental development challenges. George described that Gavin functions 3 or 4 years behind his actual age. Gavin currently attends developmental classes in high school that focus on teaching him the social skills he will need. He takes medication to treat behaviour issues that include aggressive outbursts.
- [248] Mary was a woman who was devoted to her grandchildren and was actively involved in their lives. Mary arranged for care and supports for Gavin, including community living and counselling programs to address his behavioural issues. She was described as the nurturing maternal figure for the boys. George testified that Mary was "their mother, nurse and grandmother" providing them with the emotional support that they needed.
- [249] George and Mary had an excellent relationship. George testified that Mary was his best friend and partner. He respected the way she raised her children and the way she bonded with his son, Kevin. Mary had high standards for the children. She ensured that they kept their rooms clean and did their chores. She made the family great meals and kept the home clean. She was an overall energetic and caring Mother.
- [250] When Mary underwent surgery in October, 2015, the family was not prepared to hear the news that she had cancer. George testified that the news crushed him. He

couldn't think and he couldn't function. He felt like he was losing his best friend and partner.

[251] When they told Mary about the cancer the next day, Mary was devastated. George testified that she melted inside of herself and shut down. A few days later they told the boys, who broke down and cried.

[252] Mary initially underwent treatment which included a right hemicolectomy and chemotherapy. The treatment made Mary very ill. After a few chemotherapy treatments, she decided that she wanted to enjoy the time she had left with her children and grandchildren and so stopped treatment.

[253] George described that Mary was in pain and threw up a lot. They moved her bed to the living room so that the boys would not hear her vomiting at night. Mary went from 145-150 pounds to 75 pounds or less at the time of her death.

[254] When Mary was admitted to the hospital on a palliative basis on October 14, 2016, she was very ill and suffering from intractable nausea.

[255] George described Mary's last month while in palliative care. She couldn't eat or drink anything. When she would occasionally want to taste something, they would bring her food she liked so she could chew it and then spit it out. Sometimes George could bring her home to sit on her deck and spend time with her dogs, but as time went on this become more and more difficult and Mary began to sleep more and more.

[256] The evidence at trial established that the impact of Mary's death on George and the two boys has been profound.

[257] George has been under the care of a doctor and takes medication for his depression. He also attends counselling. George has not been able to return to work because of his mental illness and the need to be at home with the boys. George testified that he anticipates returning to work but will have to find a job where he is in town consistently to take care of the boys who are now 14 and 16 years old. George testified that parenting the boys is tough as they are now in high school and he is trying to keep them on the right track.

[258] Darlene Fleury testified that she has observed the impact of Mary's death on George and the boys. She testified that George isn't the same person. He is dealing with the trauma of watching the long, drawn-out death of the love of his life. George seems under a lot of stress and pressure with the boys as he is now the sole caregiver.

[259] Darlene Fleury testified that she doesn't think the boys have gotten over Mary's death. Having been abandoned by their biological mother, they had to then watch their second mother slowly die. Darlene testified that the boys seem angry and upset sometimes and will talk about their Nana.

[260] For the purposes of calculating pecuniary damages, the court accepts that Mary and George had been a couple for 16 years at the time of Mary Fleury's death. The evidence at trial established and the court accepts that they had a loving, stable and supportive relationship.

[261] The court also accepts that George McAmmond's inability to work is directly related to the death of Mary Fleury and the emotional impact it has had on him. The court further finds that there is no evidence that George has failed to mitigate his losses in this regard.

A. Non-Pecuniary Damages – Estate of Mary Fleury – Psychological and Physical Pain and Suffering.

[262] Mary Fleury's estate is entitled to non-pecuniary damages related to the psychological and physical pain and suffering that Mary Fleury experienced as a result of the delayed diagnosis of cancer directly related to the negligence of the Defendant.

[263] The Plaintiffs have referred the court to *Adair Estate v. Hamilton Health Sciences Corp.*, 2005 CanLII 18846 (ONSCJ) – a case in which the plaintiff died as a result of a bowel obstruction. In that case the plaintiff suffered three weeks of nausea and pain, and some knowledge that she might die. The Adair Estate was awarded \$50,000 in damages (adjusted to 67,000 in 2021). For the purpose of its analysis the court has relied on the Bank of Canada calculator to adjust awards in prior cases to their value in 2021.

[264] The Plaintiffs submit that the Estate of Mary Fleury should be awarded non-pecuniary damages in the range of \$125,000 to \$150,000.

[265] The Defendant points out that for most of the period of time between October, 2011 and October, 2015, Mary Fleury did not know she had cancer and did not have physical symptoms of cancer. As such, the period of time the court should consider in assessing non-pecuniary damages is the time between Mary Fleury becoming symptomatic (July, 2015) and her ultimate death (November 26, 2016).

[266] The Defendant has referred the court to *McClintock v. Alidina*, 2011 ONSC 137 as an analogous case. In that case the plaintiff sought damages for a five-year delay in diagnosing breast cancer. The plaintiff did not die. The court in that case concluded that the treatment, recovery and prognosis would have been the same had the cancer been diagnosed earlier, however provisionally awarded general damages to the plaintiff in the amount of \$30,000 (adjusted to 35,000 in 2021).

[267] The Court agrees with the Defendant's submission that the period of time during which Mary Fleury suffered psychological and physical pain and suffering is between June, 2015, when she began to experience physical symptoms, and the date of her death in November, 2016.

[268] In addition to the cases referred to by the Plaintiffs and Defendant, the court also considered the following analogous cases:

Newman Estate v. Swales, 2002 CarswellOnt 5581 (ONSC), a delayed cancer diagnosis case where the court found that the deceased struggled with cancer treatment and knowledge that her life was coming to an end for 22 months. The Estate was awarded general, non-pecuniary damages in the amount of \$60,000 (adjusted to \$85,459 in 2021).

Stell v. Obedkoff, 2000 CarswellOnt 4085 (ONSC), a delayed cancer diagnosis case where the court found that the deceased suffered for four years prior to death. The Estate was awarded non-pecuniary damages in the amount of \$135,000 (adjusted to \$201,034 in 2021).

[269] The Court has relied on the Bank of Canada inflation calculator to adjust the damages awarded in analogous cases to their approximate value in 2021. The Court appreciates that this is an inexact method and relies on the calculator as a general guide.

[270] In this case, Mary Fleury began to exhibit symptoms of advanced disease in the summer of 2015. This resulted in a number of visits to the hospital with complaints of pain in her right lower quadrant. This led to an endoscopy being performed in September, 2015, and a laparoscopy on October 31, 2015.

[271] The evidence at trial is that Mary Fleury would have likely had a right hemicolectomy and adjunctive chemotherapy if she had been properly diagnosed in 2011. As such, she would have experienced pain and suffering related to these procedures regardless of the delayed diagnosis.

[272] However, Mary Fleury would not have suffered from the advanced illness that presented in 2015. This included a bowel obstruction, and persistent nausea and vomiting that continued after chemotherapy was discontinued.

[273] Mary Fleury would not have had to live for over a year with the knowledge that she was going to die and that she would be deprived of precious time with her spouse, her children, her grandchildren, family and friends. Further, she would not have suffered the long, drawn-out death that she experienced in the fall of 2016.

[274] Having regard to the analogous cases, and the particular circumstances in this case, the court concludes that the Estate of Mary Fleury is entitled to non-pecuniary damages for psychological and physical pain and suffering in the amount of **\$120,000**.

B. Non-Pecuniary Damages – Family Law Act Claim – Damages for Loss of Care, Guidance and Companionship

[275] Section 61 of the *Family Law Act*, R.S.O. 1990, c.F-3 provides as follows:

61.(1) If a person is injured or killed by the fault or neglect of another under circumstances where the person is entitled to recover damages, or would have been entitled if not killed, the spouse, as defined in Part III (Support Obligations), same-sex partner, as defined in Part III (Support Obligations), children, grandchildren, parents, grandparents, brothers and sisters of the person are entitled to recover their pecuniary loss resulting from the injury or death from the person from whom the person injured or killed is entitled to recover or would have been entitled if not killed, and to maintain an action for the purpose in a court of competent jurisdiction.

Damages in case of injury

(2) The damages recoverable in a claim under subsection (1) may include,

(a) actual expenses reasonable incurred for the benefit of the person injured or killed;

(b) actual funeral expenses reasonably incurred;

(c) a reasonable allowance for travel expenses actually incurred in visiting the person during his or her treatment or recovery;

(d) where, as a result of the injury, the claimant provides nursing, housekeeping or other services for the person, a reasonable allowance for loss of income or the value of the services; and

(e) an amount to compensate for the loss of guidance, care and companionship that the claimant might reasonably have expected to receive from the person if the injury or death had not occurred.

[276] Courts have struggled to quantify the amount of money that can reflect the loss of a loved one. As Justice Nordheimer stated in *Hechavarria v. Reale* [2000] O.J. No. 4288 (ONSCJ) at para. 11:

As anyone will quickly appreciate, and certainly any judge that has to undertake the task, it is impossible to quantify the loss of life in monetary terms. No amount of money could ever replace the contribution which a loved one makes to the people around him or her. The loss of the care, companionship, guidance and the very

important emotional bond cannot be calculated in dollar terms. The awarding of money, however, is the only remedy which a court has available to it in circumstances such as this. Further, when considering the amount to be awarded, a judge must be constrained to a range that is consistent with awards previously made in other cases so that the results in any given case will have a measure of consistency with other like cases. In that way, the courts try to achieve a degree of predictability which in turn, it is hoped, will assist future litigants to resolve such matters without the need for a full trial with all of its attendant hardships and expense.

[277] What “loss of care, guidance and companionship” was described by the Ontario Court of Appeal in *To v. Toronto Board of Education*, (2001), 2004 D.L.R. (4th) 704 at para 36 as follows:

Companionship, as it was defined in *Mason v. Peters* in a fatal accident context, consists of the deprivation of the society, comfort and protection which might reasonably be expected had the child lived. Robins J.A. described it as “the loss of the rewards of association which flow from the family relationship”. Care was referred to by Linden J. in *Thornborrow v. MacKinnon* (1981), 32 O.R. (2d) 740 (Ont. H.C.) as including “feeding, clothing, cleaning, transporting, helping and protecting another person”. *Thornborrow* was cited with approval by Robins J.A. in *Mason v. Peters*. See also *Huggins v. Ramtej* (Ont. S.C.J.). In *Thornborrow*, Linden J. described guidance as including such things as education, training, discipline and moral teaching.

[278] Each case depends on a full assessment of the evidence, on a case-by-case basis. Various factors to be considered include the circumstances of the death, the age of the deceased, the age of dependent, the nature and quality of the relationship between the deceased and the dependent, the dependent’s personality and ability to manage the emotional consequences of the death, and the effect of the death on the dependent’s life. *Wilcox v. “Miss Megan” (The)*, 2007 CarswellNat3324 (FC) at para 90.

[279] This assessment must be objective, unemotional and based on the evidence presented to the court. The assessment does not include compensation for the grief, sorrow and mental anguish suffered. *Lawrence Estate v. Alexander*, 2009 CarswellOnt 1148 (ONSCJ) at para. 502; *Reidy v. McLeod Estate*, 29186 CarswellOnt 768 (ONCA) at para. 4.

George McAmmond

[280] George McAmmond testified about the impact Mary Fleury’s death has had on him. Her diagnosis and death “crushed” him. Since her death he has had trouble

functioning and has suffered from mental illness. He is under the care of a doctor, who has prescribed medication for his depression. He continues to attend counselling.

[281] In addition, George has lost the guidance of Mary when it comes to raising her two grandchildren, Damian and Gavin, who are now teenagers. The evidence at trial established, and the court accepts, that Mary was very much a “hands on” caregiver for the two boys, and in particular with respect to supporting Gavin’s special needs.

[282] The court accepts that George has struggled with the responsibility of raising the boys alone and has experienced increased stress as a result of the loss of guidance and support from Mary.

Damian Reid and Gavin Bradford

[283] With respect to Damian and Gavin, the court accepts that the loss of their grandmother, who for all intents and purposes was their mother, was devastating to them. The court accepts that it would be extremely difficult to lose Mary, after having been abandoned by their mother. George McAmmond testified that Mary was their “mother, nurse and grandmother”. She was the warm and nurturing presence in their lives.

[284] The court also accepts that Gavin received particular care and guidance from Mary in relation to his special needs. This included ensuring Gavin received access to appropriate community supports. The court accepts that Gavin would have needed supports from Mary well into his early adulthood.

Assessment of Non-Pecuniary Damages

[285] With respect to the appropriate damages for George McAmmond, for loss of care, guidance and companionship, the Plaintiffs submit the appropriate range would be between \$150,000 and \$200,000.

[286] The Plaintiffs point to the Ontario Court of Appeal decision in *Moore v. 7595611 Ontario Ltd*, 2021 ONCA 459, where the Court upheld a jury award of \$250,000 each to both parents of a child who died as a result of a fire. In that case the parents had to watch their daughter suffer in hospital for several days before her death.

[287] The Defendant submits that the appropriate amount to be awarded to George McAmmond would be \$50,000 and relies on *Tahir v. Mitoff*, 2019 ONSC 7298 as an analogous medical negligence case where the husband of the deceased received \$50,000.

[288] The Plaintiffs submit that the appropriate amount of damages to compensate Damian Reid would be \$100,000 to \$125,000. The Plaintiffs submit that the appropriate amount of damages to compensate Gavin Bradford would be \$150,000 to \$175,000 given his special needs.

[289] The Defendant submits that the appropriate amount to be awarded to Damian Reid and Gavin Bradford would be \$35,000 each.

[290] The Defendant relies on the following cases as analogous:

Tahir v. Mitoff, sura., where the adult children received \$35,000;

Adair Estate v. Hamilton Health Services Corp, supra, where the children were awarded \$20,000 each (adjusted to 25,800 in 2021); and

McVea (Guardian ad litem of) v. B(T) Guardian ad item of, 2002 BCSC 1407 where two young children (ages 3 and 5) were awarded \$30,000 each (adjusted to \$42,000 in 2021).

[291] The Court has examined other analogous cases that relate to damages awarded to spouses and children for loss of guidance, care and companionship, and has considered the following additional cases:

1. *Hechavarria v. Real*, 2000 CarswellOnt 4099 (ONSC) – where a 53 year old mother was killed in a motor vehicle collision. The husband was awarded \$85,000 (adjusted to \$126,577 in 2021), and the adult children were awarded \$30,000 (adjusted to 44,674 in 2021);
2. *Fish v. Shainhouse*, [2005] O.J. No. 4575 (ONSC) – where the deceased died two days after presenting at the hospital with chest pains. Husband was awarded \$80,000 (adjusted to 107,063 in 2021), a dependant child, aged 13, was awarded \$80,000 (adjusted to 107, 063 in 2021), and an adult daughter was awarded \$50,000 (adjusted to 66,914) in 2021);
3. *Stell v. Obedkoff*, 2000 CarswellOnt 4085 (ONSC) – where there was a delayed diagnosis of cancer. Deceased died at age 46. The husband was awarded \$50,000 (adjusted to \$74,457 in 2021) and the children were awarded \$20,000 (adjusted to \$29,782 in 2021);
4. *Wilcox v. “Miss Megan”(The)*, 2007 FC 1004 (FCJ) – where the deceased died in a boating accident. The wife was awarded \$75,000 (adjusted to \$96,428 in 2021) and an adult disabled child received \$75,000; and
5. *Nielsen v. Kaufman*, 1986 CarswellOnt 764 (ONCA). Two children, ages 12 and 6 at the time of their mother’s death. Elder son received \$20,000 (adjusted to \$43,113 in 2021) and younger son received \$30,000 (adjusted to \$64,670).

[292] Having reviewed the circumstances in this case, and having considered the analogous cases, the Court concludes that the following damages are appropriate for the loss of Mary Fleury's care, guidance and companionship: George McAmmond – **\$100,000**; Damian Reid - **\$50,000** and Gavin Bradford- **\$65,000**.

C. Pecuniary Damages:

[293] The basic principle of compensation for pecuniary damages is to put the Plaintiffs in the economic position they would have been in, but for the negligence of the defendant. *Andrews v. Grand & Toy Alberta Ltd.*, [1978] 2 S.C.R. 229.

[294] The standard of proof that the Plaintiffs must meet for future losses is the standard of a "real and substantial possibility" of the future pecuniary loss claimed. *Schrump v. Koot*, 1977 CanLII 1332 (ONCA).

[295] The court has been provided with the expert evidence of two economists, Dr. Eli Katz on behalf of the Plaintiffs and Dr. Douglas Hyatt on behalf of the Defendant. The experts are not significantly far apart in their opinions with respect to the pecuniary damages in this case, indeed their final calculations are only \$74,103.00 apart.

[296] The experts differ on whether the court should apply the modified sole dependency approach (Dr. Katz) or the cross-dependency approach (Dr. Hyatt) to assessing the quantum of damages.

[297] Both experts were well qualified and provided their evidence in a clear and convincing manner. As such, the court accepts the accuracy of their calculations and underlying factual assumptions.

[298] The court concludes that the sole issue to be determined in this case, with respect to the issue of pecuniary damages, is whether the court should accept the modified sole dependency approach or cross-dependency approach when measuring the damages.

Dr. Eli Katz

[299] Dr. Eli Katz was called by the Plaintiffs. He testified that there are two measurements used by economists to quantify dependency loss. The first is the sole dependency measure which measures the amount of benefit that the surviving spouse and child received from the income of the deceased. This is normally calculated at a rate of 70% for the spouse and 4% for each child.

[300] Dr. Katz testified that the second measurement used by economists is the cross-dependency measure, which considers that the amount of loss should be reduced by

an amount that reflects the fact that the deceased would have used some of their income to their own benefit.

[301] Dr. Katz testified that it was his opinion that a “modified sole dependency” measure was appropriate in this case. It was his opinion that most families are a combination of sole dependency and cross dependency, as such the middle ground is to apply a 60% dependency rate. In essence, that as a result of her death, George McAmmond lost 60% of Mary Fleury’s net income, and each child lost 4% of her income.

[302] Included in his calculations were allowances for various contingencies such as inflation, survival rates, retirement, divorce, average contribution towards housework, and the age at which child dependency ends. During his evidence, Dr. Katz testified that he took a very conservative approach when applying these contingencies, at the request of the Plaintiffs.

[303] Dr. Katz calculated the value of loss of household services due to Mary Fleury’s death. In reaching his conclusion, Dr. Katz discounted the amount to reflect that some of the household services that Mary Fleury would have provided would have been for her own benefit. Dr. Katz discounted the amount by 10% to reflect this.

[304] The Plaintiffs asked Dr. Katz to calculate damages up to the date of March 20, 2020, which corresponds to the date on which the world-wide Covid 19 pandemic brought much of the world to a standstill with various lock downs.

[305] The Plaintiffs ask this court to consider awarding damages post March 20, 2020 on the basis of the methodology used by Dr. Katz. In this case, the court declines to do so without a proper factual foundation. For example, there is no evidence before the Court addressing whether George McAmmond received CERB benefits provided by the Canadian government during 2020 and 2021 and whether those benefits should be deducted from the approximate annual income loss due to his inability to work and his annual income for purposes of determining the percentage past loss of financial support. Had the Plaintiffs wanted the court to make this adjustment, they should have called evidence with respect to this time period and had Dr. Katz provide evidence regarding this issue to the court.

[306] The Plaintiffs also ask this court to consider adjusting Dr. Katz’s calculations upwards to account for the particular evidence in this case with respect to Mary Fleury’s contribution to household work, the unlikelihood of separation given the strength of George and Mary’s relationship, the likelihood that Mary would have worked past 60 years of age, the fact that Damian and Gavin are teenagers and thus likely to consume more of the household income, and Gavin’s special needs and the impact on the length of his dependency.

[307] The court finds that it is unable to make sufficient findings of fact, based on the evidence led at trial, that would allow it to adjust the amounts provided by Dr. Katz upwards. Again, had the Plaintiffs wished to the court to make this adjustment, they

could have led evidence through Dr. Katz with respect to other possible calculations; they did not.

Dr. Douglas Hyatt:

[308] Dr. Douglas Hyatt, who was called by the Defendant, testified that the cross-dependency approach determines the amount required by the surviving spouse to maintain the same standard of living as when the deceased spouse was alive. This measurement considers that the surviving spouse is no longer spending portions of their income on the deceased spouse. To determine this amount, the economist calculates 72% of the total family income and then reduces that amount by the surviving spouse's ongoing income.

[309] Dr. Hyatt testified that the cross-dependency approach is supported by statistical data that shows that 72% is the amount required by a dependent spouse to maintain the same standard of living as they would have enjoyed had the deceased party still been alive.

[310] With respect to the need to deduct an amount to reflect how much of those services would have been for Ms. Fleury's own benefit, Dr. Hyatt relied on a study by Cara Brown, that found that the dependency rate in wrongful death cases ranged from 50-90%, depending on the division of household labour, loss of economies of scale and the reduction of household labour required in the absence of the deceased. Using this study, Dr. Hyatt used a dependency rate of 80%.

[311] Dr. Hyatt was asked by the Defendant to assume three scenarios: firstly, that George's withdrawal from employment was not causally connected to the negligence of Dr. Kassim (scenario A); secondly, that George's withdrawal from employment was causally connected to the negligence of Dr. Kassim (scenario B) and thirdly, that George's withdrawal from employment was causally connected to the negligence of Dr. Kassim, however George failed to mitigate his losses (scenario C).

[312] The evidence at trial did not support either scenario A or C. As such, the court turns to scenario B – which was supported by the court's finding of fact that George's inability to work is causally connected to the defendant's negligence and there has been no evidence of a failure by him to mitigate those circumstances.

Analysis:

[313] Determining which approach to apply in this case is not an easy task as neither fits perfectly into the facts in this case. Having reviewed the evidence of Dr. Katz and Dr. Hyatt, the court concludes that the approach that most closely fits the facts in this case is the cross-dependency approach.

[314] In support of this conclusion the court relies on the evidence of George McAmmond who testified that he and Mary approached their contributions to the family equally. They pooled their resources and split the bills 50/50. Although Mary contributed to the household work at a higher rate than George, George contributed to outdoor maintenance and handyman work at a higher rate than Mary. Finally, as pointed out by Dr. Hyatt, George McAmmond's salary was significantly higher than Mary Fleury's (almost double) which does not support a sole dependency approach.

[315] The court adopts scenario B, as outlined in Dr. Hyatt's report which reflects the fact that George's inability to work was causally connected to the defendant's negligence.

[316] The court adopts the valuation date of December 9, 2021, which was adopted by both Dr. Katz and Dr. Hyatt for the calculation of pecuniary damages.

[317] Having considered the totality of evidence called at trial, and applying the approach outlined by Dr. Hyatt, the court makes the following findings with respect to damages:

1. The court concludes that the **past loss of earnings of George McAmmond** amounts to **\$287,018**. Both the Plaintiffs and Defendant agree to that this amount reflects his loss of income between November 26, 2016 and December 8, 2021 based on his income tax returns and accepting that his withdrawal from paid employment was causally related to the Defendant's negligence.
2. The court concludes that the **past loss of financial support to George McAmmond** amounts to **\$33,153**. This amount reflects the period from November 26, 2016 to December 8, 2021 and is based on Ms. Fleury's Net Income for the years 2016 to 2020 as reflected in her income tax returns. The court accepts Dr. Hyatt's evidence that the annual loss of financial support would equal 72% (after-tax) household income, less the continuing (after-tax) income of Mr. McAmmond. The court accepts Dr. Hyatt's calculation that this would amount to \$33,153.
3. The court concludes that the **past loss of financial support to Damian and Gavin** amounts to **\$4,912 each**. Both Dr. Katz and Dr. Hyatt approach this calculation in the same manner, calculating 4% of Mary Fleury's net income, per child. Both calculated this to be \$4,912 each, which the court accepts as accurate.
4. The court concludes that **future loss of financial support to George McAmmond** amounts to **\$190,028**. This amount is premised on the assumption that Mary Fleury would have had a normal life expectancy and uses the same calculation for past loss of financial support, 72% (after-tax) household income, less the continuing (after-tax) income. The court

accepts Dr. Hyatt's calculation, based on the cross-dependency approach, that this number would be \$190,028.

5. The court concludes that the **future loss of financial support to Damian Reid** amounts to **\$6,067**. The court accepts that Damian Reid would likely be dependent on Mary Fleury's earnings until the age of 22. This calculation is based on 4% of Mary Fleury's net earnings. The court accepts Dr. Hyatt's calculation that this would amount to \$6,067.
6. The court concludes that the **future loss of financial support to Gavin Bradford** amounts to **\$10,006**. The court accepts that Gavin Reid would likely be dependent on Mary Fleury's earnings until the age of 25, given his special needs. The court accepts Dr. Hyatt's calculation that this would amount to \$10,006.
7. The court concludes that the **past loss of household services** amounts to \$103,104, and the **future loss of household services amounts** to \$413,983. As such, the total loss of household services amounts to **\$517,087**. This assumes a dependency rate for household services of 80%. The court accepts Dr. Hyatt's calculations with respect to these amounts.

[318] As such, the total amount of pecuniary losses arising from the death of Mary Fleury are set at **\$1,053,183**.

[319] The following table summarizes the court's conclusion with respect to pecuniary damages:

George McAmmond past loss of earnings	\$287,018
Past loss of financial support to George McAmmond	\$33,153
Past loss of financial support to Damian Reid	\$4,912
Past loss of financial support to Gavin Bradford	\$4,912
Future loss of financial support to George McAmmond	\$190,028
Future loss of financial support to Damian Reid	\$6,067

Future loss of financial support to Gavin Bradford	\$10,006
Past loss of household services	\$103,104
Future loss of household services	\$413,983
Total:	\$1,053,183

Out of pocket expenses

[320] The parties agree that actual out of pocket funeral expenses incurred in this case amounted to **\$5,327.35**.

OHIP

[321] The parties also agree that OHIP's entire subrogated claim in this matter amounts to **\$46,232.85**, representing the cost of Mary Fleury's palliative care.

D. Pre-judgement interest:

[322] The Defendant submits that while pre-judgement interest runs from the date of the cause of action to the date of the order, the appropriate date for any interest to run in this case is from the fall of 2015 when Mary Fleury became aware that she had cancer and not from October, 2011 when the tort of negligence occurred.

[323] The Defendant further submits that this court should exercise its discretion pursuant to section 130 of the *Courts of Justice Act*, R.S.O. 1990, c.C.43 ("CJA") to apply a lower rate of pre-judgement interest than the 5% as prescribed in Rule 53.10 of the *Rules of Civil Procedure*, R.R.O. 1990, Reg. 194 ("the Rules"). The Defendant refers to the Ontario Court of Appeal decision in *MacLeod v. Marshall*, 2019 ONCA 842 in support of its submission.

[324] The Plaintiffs submit that *MacLeod v. Marshall* does not set the law for all cases. Rather, that decision reflects the fact that the claim against the Defendants in that case spanned over 30 years. The Plaintiffs submit there is no reason in this case to deviate from Rule 53.10 of the *Rules* which remains in force.

[325] Section 130 of the CJA provides a trial judge with discretion, where it considers it just to do so, to depart from the default rate. The factors the court is to consider in exercising its discretion are found in section 130(2) and are as follows:

Section 130 (2)

- (a) changes in market interest rates;
- (b) the circumstances of the case;
- (c) the fact that an advance payment was made;
- (d) the circumstances of medical disclosure by the plaintiff;
- (e) the amount claimed and the amount recovered in the proceedings;
- (f) the conduct of any party that tended to shorten or to lengthen unnecessarily the duration of the proceeding; and
- (g) any other relevant consideration.

[326] In this case the court agrees that the pre-judgement interest should begin in 2015, when Mary Fleury began to experience symptoms likely related to appendiceal cancer which the court finds was in June, 2015. For the purposes of calculation, the pre-judgement interest should begin on June 1, 2015.

[327] Having considered the factors outlined in section 130(2), the court notes that in this case the period of pre-judgment interest will run from June 1, 2015 to the date of judgement, a period of approximately 7 years. This is not a case where the pre-judgement interest will apply for an excessively long period of time.

[328] During the time period between June, 2015 and December, 2021, the average rate of inflation fluctuated between 1.0% and 4.8%. The court takes judicial notice that in March, 2020 much of the world shut down due to an unprecedented global pandemic. The impact of the global pandemic on the world economy continues to today's date and the rate of interest continues to increase. This adds a level of uncertainty to any analysis with respect to pre-judgement interest.

[329] Having considered the changes in market interest rates, the impact of a global pandemic, and the circumstances of this case the court declines to exercise its discretion to depart from the pre-judgement interest rate for non-pecuniary damages as set out in Rule 53.10. Prejudgment interest on non-pecuniary damages will therefore accrue at a rate of 5% per annum, commencing on June 1, 2015.

Exhibits

[330] During the course of the trial an issue arose with respect to whether exhibits should be numbered or lettered. Ultimately the issue was resolved by way of an agreement between the parties that those materials referred to by the witnesses as part of their evidence at trial would be made numbered exhibits. For clarification, all exhibits that were numbered form part of the trial proper. Exhibits that remained lettered did not become part of the trial proper. A copy of the final exhibit list is attached to this decision.

A handwritten signature in blue ink, appearing to read "Susan Stothart". The signature is stylized with a large, sweeping initial 'S'.

The Honourable Madam Justice S. K. Stothart

Released: April 22, 2022

THE ESTATE OF MARY FLEURY et al v. OLAYIWOLA A. KASSIM 2022 ONSC 2464
COURT FILE NO.: CV-16-084
DATE: 2022/04/22

BETWEEN:

RE: THE ESTATE OF MARY FLEURY, by Estate
Trustee, GEORGE MCAMMOND, GEORGE
MCAMMOND, DAMIAN REID, a minor by his
litigation guardian, GEORGE MCAMMOND, and
GAVIN BRADFORD, a minor by his litigation
guardian, GEORGE MCAMMOND

– and –

OLAYIWOLA A. KASSIM

Released: April 22, 2022